



Deepfake Detection

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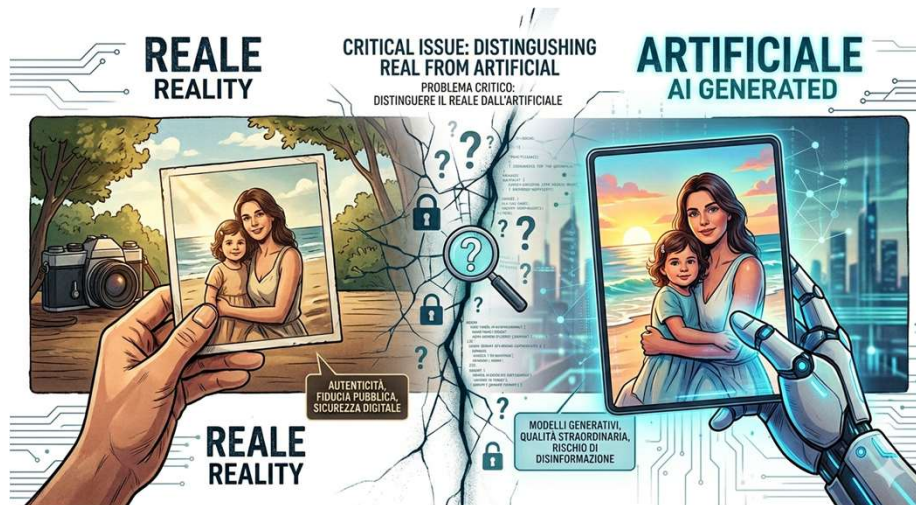
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MATEMATICA
E INFORMATICA

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La Sfida: Contenuti Sintetici Sempre Più Sofisticati

I modelli generativi producono oggi contenuti visivi e audio di qualità straordinaria. Distinguere il reale dall'artificiale è diventato un problema critico per la sicurezza digitale, l'informazione e la fiducia pubblica.



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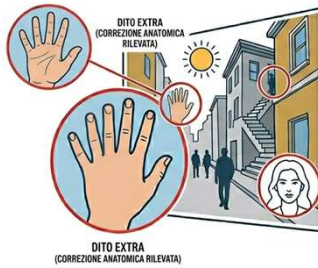
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Come Riconoscere un Deepfake?

- Il primo livello di difesa è l'analisi visiva e semantica. Comprendere dove i modelli generativi falliscono è il punto di partenza per costruire sistemi di rilevamento efficaci.

DITO EXTRA (ERRORE GEOMETRICO)



ANALISI VISIVA

ANALISI SEMANTICA: OLTRE LA SUPERFICIE VISIVA

LOGICA DELLA SCENA FALLITA



ANALISI SEMANTICA

OBIETTIVO: VERIFICARE LA VERIDICITÀ LOGICA E CONTESTUALE, NON SOLO VISIVA



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L'Ispezione Visiva: Indizi Sottili

→ Artefatti Visivi

Anomalie cromatiche, riflessi corneali innaturali, bordi sfumati intorno a capelli e orecchie.

→ Artefatti Semantici

Asimmetrie facciali, incongruenze tra luce e ombre, denti o dita malformati.



ANALISI DEI GENERATIVE ARTEFATTI

OBIETTIVO: IDENTIFICARE E CLASSIFICARE TUTTI GLI ARTEFATTI VISIBILI E SEMANTICI PER VERIFICARE L'ORIGINE DELL'IMMAGINE, NON SOLO LA SUA LOGICA



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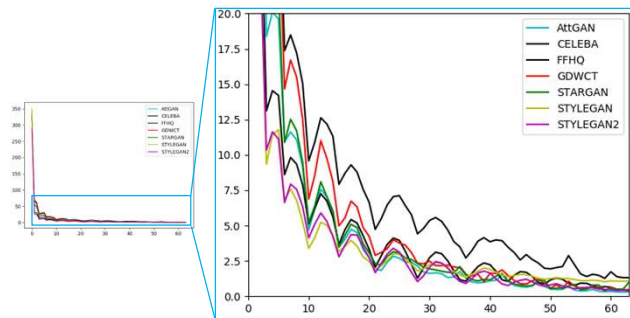
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Oltre l'Occhio Umano: L'Analisi Tecnica

Artefatti di Basso Livello

Ogni modello generativo lascia un'impronta nel rumore dell'immagine, rilevabile con analisi statistica.



Dominio della Frequenza

Lo spettro di potenza rivela differenze sistematiche tra immagini reali e sintetiche.

Le immagini generate da AI, in generale, mostrano picchi di frequenza regolari e pattern periodici assenti nelle immagini reali. Questa firma spettrale è uno degli indicatori (traccia) per la deepfake detection.

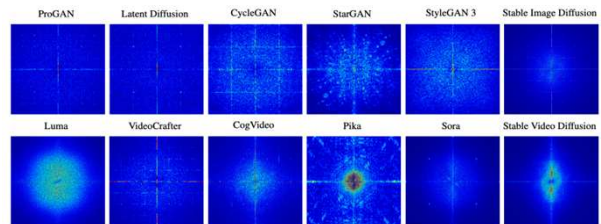
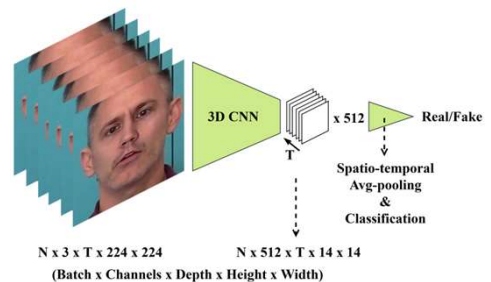
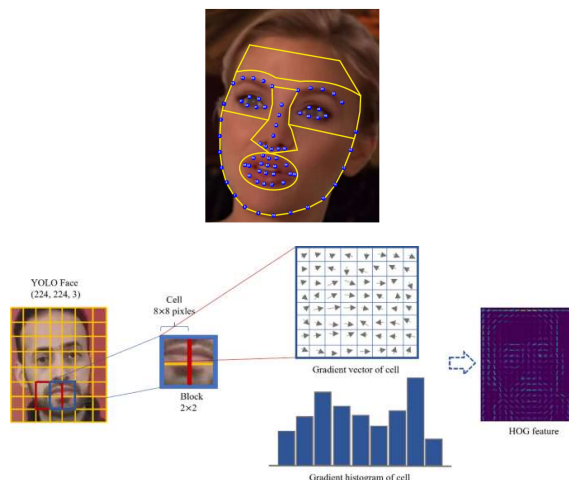


Figure 3. Fourier transform analysis of the forensic traces extracted from different synthetic image and video generators.

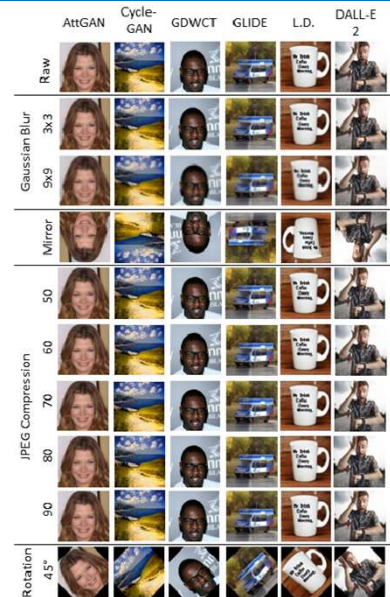
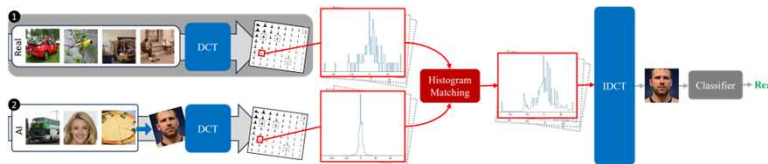
Tecniche di Deepfake Detection

Le metodologie più avanzate combinano deep learning, analisi forense del segnale e modellazione spazio-temporale per identificare manipolazioni invisibili all'occhio umano.



Obiettivi di un Deepfake Detector

- Robustezza ad attacchi
 - JPEG Compression (al variare del QF)
 - JPEG AI (al variare del bpp)
 - Scaling
 - Gaussian Blur
 - Mirroring
 - Upload/Download da Social Network
 -
 - Attacchi Adversariali



I limiti dei Deepfake Detector

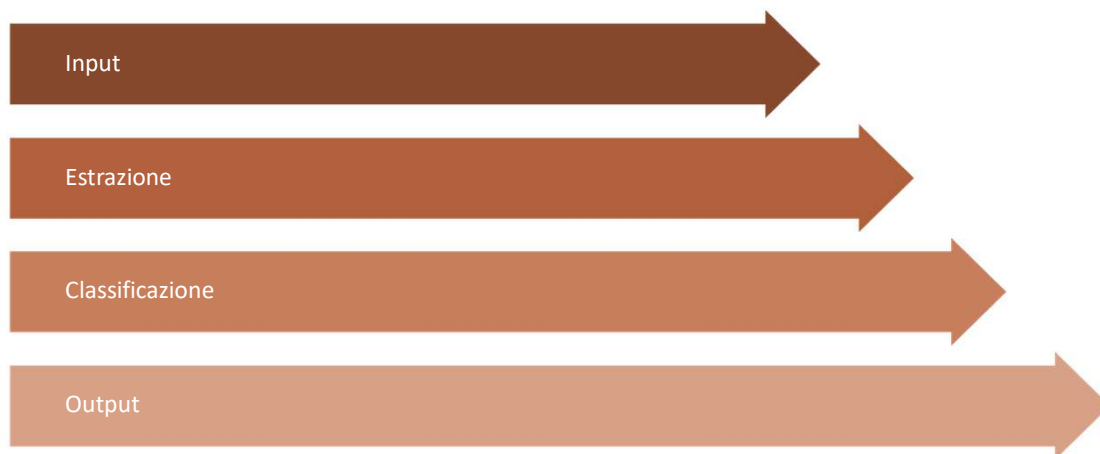
L'obiettivo è la generalizzazione

- Un detector deve funzionare su modelli generativi che non ha mai visto durante il training.



Generica Pipeline di Deepfake Detection

Una pipeline tipica di detection analizza l'input a più livelli: dal pixel grezzo alle relazioni strutturali tra regioni facciali.



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Deepfake Detector: Categorie

- Il rilevamento dei **deepfake** (deepfake detection) si avvale di due categorie principali di approcci, spesso combinati per una maggiore efficacia: **attivi** e **passivi**.

Approccio Attivo

Questo metodo si concentra sull'inserimento di "segnali" o "impronte digitali" nel contenuto originale al momento della creazione (come watermark digitali o firme biometriche), in modo che qualsiasi alterazione possa essere facilmente individuata in seguito.

È un **approccio proattivo** che interviene prima che il deepfake venga creato.

Approccio Passivo

Questo metodo analizza i contenuti multimediali *dopo* che sono stati creati, alla ricerca di **inconsistenze** o **artefatti lasciati dai modelli generativi**.

Obiettivo: cercare artefatti a livello di pixel (come bordi sfocati o texture innaturali) o incongruenze semantiche (come movimenti oculari non sincronizzati o espressioni facciali anomale).

È un **approccio reattivo** che lavora sui contenuti esistenti.



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Deepfake Detectors: Approcci attivi e passivi

Approcci Attivi

Questi approcci richiedono l'inserimento di metadati o segnali nel contenuto al momento della creazione, per certificarne l'autenticità e tracciarne l'origine.

- **Filigrana Digitale (Digital Watermarking):** Inserimento di un codice impercettibile nel file video/audio che viene alterato se il file viene manipolato.
- **Standard di Provenienza (C2PA):** Tecniche che tracciano l'origine del file e certificano le modifiche subite lungo la catena di produzione, come la Coalizione per la Provenienza e l'Autenticità dei Contenuti.
- **Firme crittografiche:** Firma digitale del contenuto all'acquisizione, in modo che qualsiasi modifica ne invalidi la firma

Approcci Passivi

Questi approcci analizzano il contenuto multimediale dopo che è stato creato, alla ricerca di artefatti o anomalie inosservabili a occhio nudo.

- **Analisi basata su artefatti biologici:** Cerca incoerenze nei movimenti naturali, come l'assenza di battito di ciglia, movimenti oculari innaturali o deformazioni anomale della pelle.
- **Analisi del flusso ottico:** Esamina come gli oggetti si muovono tra un frame e l'altro per individuare incongruenze fisiche che il software di generazione non è riuscito a replicare perfettamente.
- **Analisi della coerenza facciale:** Verifica la corrispondenza tra espressioni facciali, movimento della bocca (lip-sync) e l'audio, poiché i deepfake spesso presentano disallineamenti.
- **Analisi dei pixel e dell'IA:** Utilizza modelli di apprendimento profondo (CNN) per rilevare tracce digitali lasciate dai modelli generativi (come GAN o Diffusione) nella struttura del file.
- **Rilevamento in tempo reale:** Soluzioni che analizzano la "vivacità" (liveness detection) per verificare che la persona davanti alla telecamera sia reale.



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Un esempio di «approccio attivo»



Aneja, Shivangi, Lev Markhasin, and Matthias Nießner. "Tafim: Targeted adversarial attacks against facial image manipulations." *European Conference on Computer Vision*. Cham: Springer Nature Switzerland, 2022.

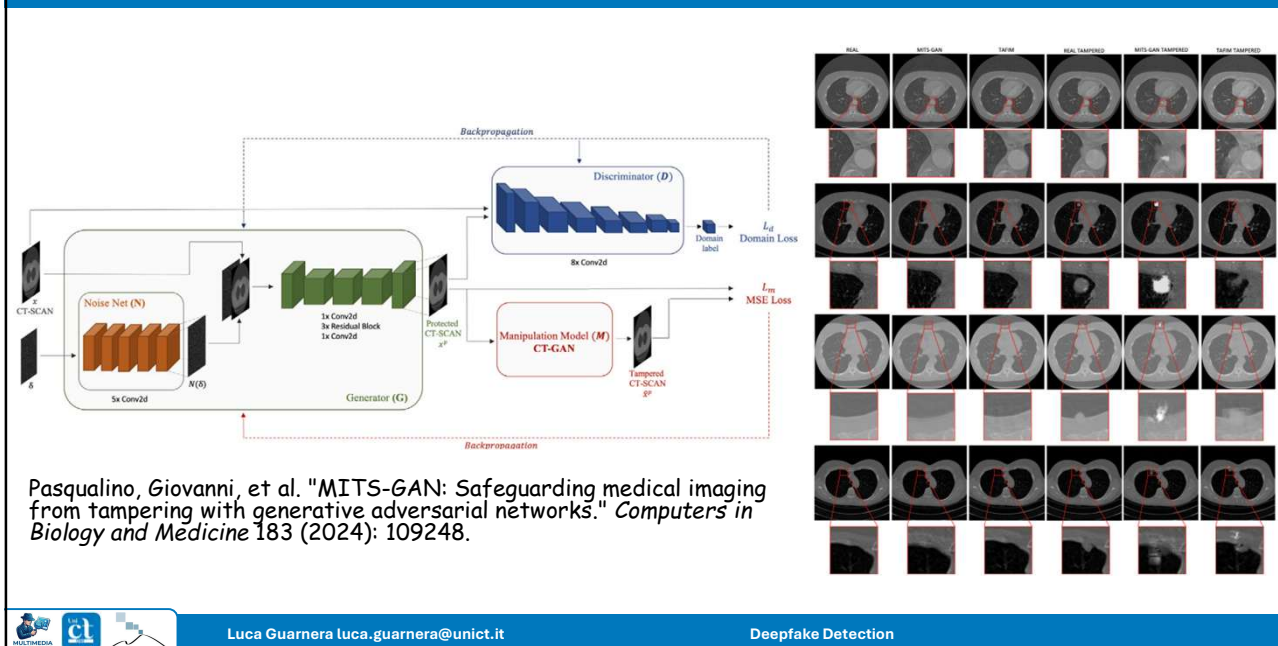


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Un esempio di «approccio attivo» su Medical Deepfake Domain



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Deepfake Detectors: «Altre dimensioni di classificazione»

Per modalità analizzata

- **Visual-only** - analisi del solo frame video o immagine
- **Audio-only** - analisi della voce (clonazione vocale, TTS)
- **Audio-visual** - sfruttano l'inconsistenza tra labiale e audio (es. lip-sync detection)
- **Multimodal** - combinano più segnali

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Deepfake Detectors: «Altre dimensioni di classificazione»

Per dominio di analisi

- **Spaziale (pixel domain)** - analisi di artefatti visivi, blending boundaries, asimmetrie facciali
- **Frequenziale** - analisi FFT/DCT per artefatti GAN tipicamente visibili nello spettro delle frequenze
- **Temporale** - analisi della coerenza tra frame consecutivi (battito delle palpebre, micro-movimenti)
- **Compressione** - artefatti JPEG/H.264 inconsistenti con la generazione sintetica



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Deepfake Detectors: «Altre dimensioni di classificazione»

Per architettura del modello

- **CNN-based** - classificatori tradizionali su features visive
- **Transformer/ViT-based** - attention su patch dell'immagine
- **Recurrent (LSTM/GRU)** - per la dimensione temporale nei video
- **Graph Neural Network** - relazioni tra landmark facciali



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Deepfake Detectors: «Altre dimensioni di classificazione»

Per tipo di manipolazione rilevata

- **Face swap** (es. FaceSwap, DeepFaceLab)
- **Face reenactment** (es. First Order Motion)
- **Face synthesis** (volti completamente GAN-generati, es. StyleGAN)
- **Attribute manipulation** (es. invecchiamento, cambio espressione)
- **Speech/voice cloning**



Deepfake Detectors: «Altre dimensioni di classificazione»

Per generalizzazione

- **In-dataset** - addestrati e testati sullo stesso dataset (alta accuracy, bassa utilità reale)
- **Cross-dataset** - testati su manipolazioni non viste in training (problema aperto)
- **Open-set / zero-shot** - tentano di rilevare manipolazioni mai viste

Per interpretabilità

- **Black-box** - output binario o score senza spiegazione
- **Explainable (XAI)** - mappe di saliency, attention visualization, spiegazioni localizzate



Deepfake Detectors

- Possiamo considerare due categorie principali:

Deepfake Detector

Approcci basati su
Machine Learning

Approcci basati su
Deep Learning



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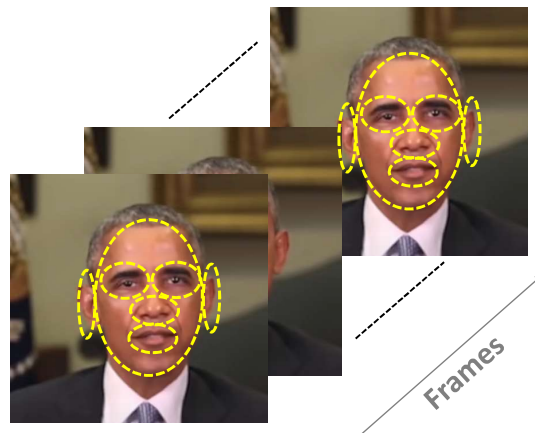
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Deepfake detection methods

Spatial Artifacts



Spatial and Temporal Artifacts



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Obiettivi

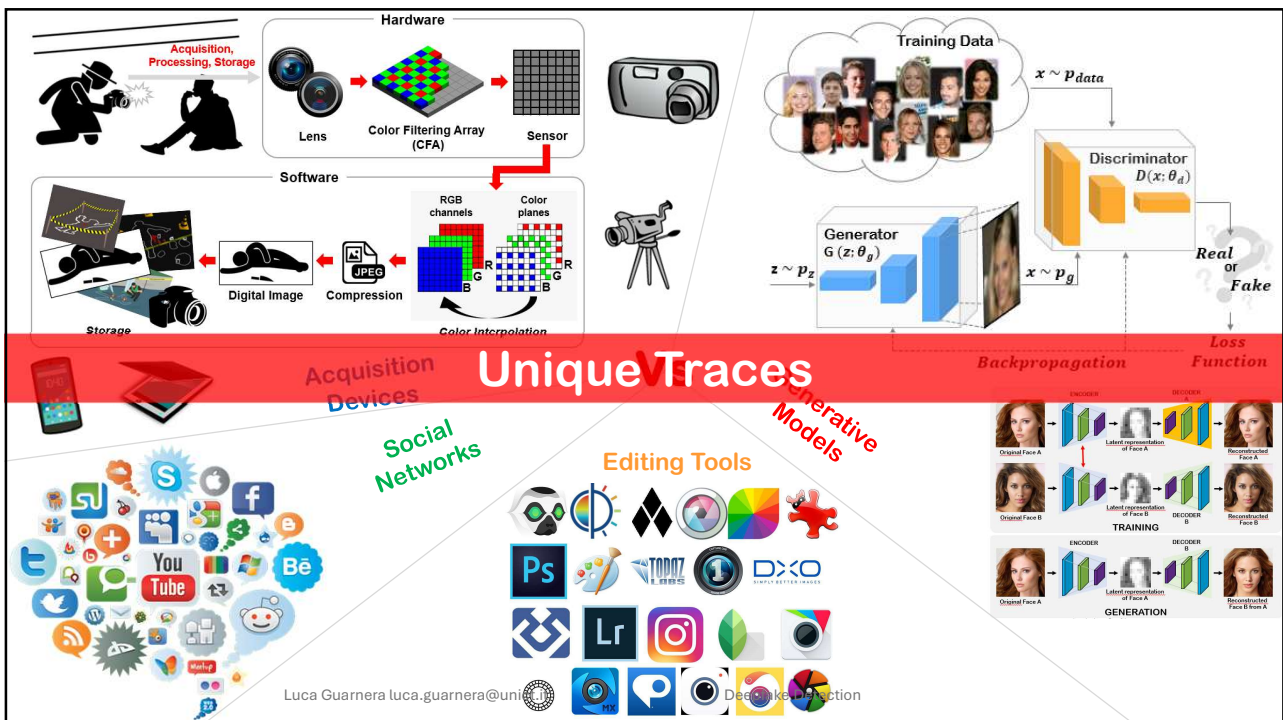
- Alla ricerca delle tracce ...
- Perché? Partiamo da un principio base ...



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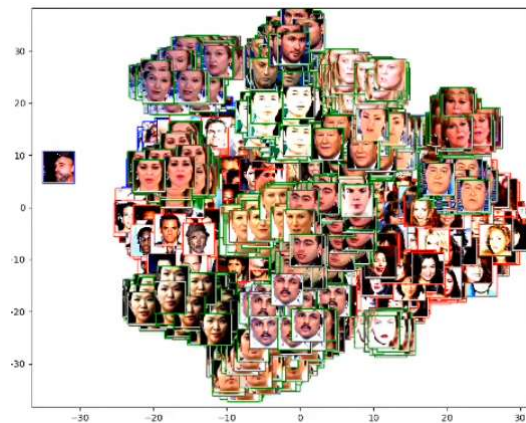


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Detecting fakes

Different GANs have different distributions ...

... Different Traces



Blue: StyleGAN
Red: ProGAN
Green: Face2Face



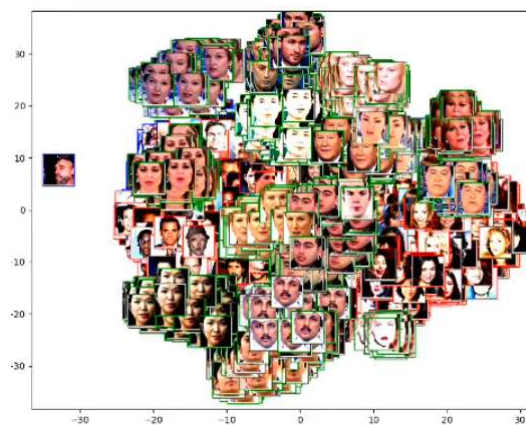
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Detecting fakes

Detectors trained on a single GAN do not generalize well on other GANs



Blue: StyleGAN
Red: ProGAN
Green: Face2Face



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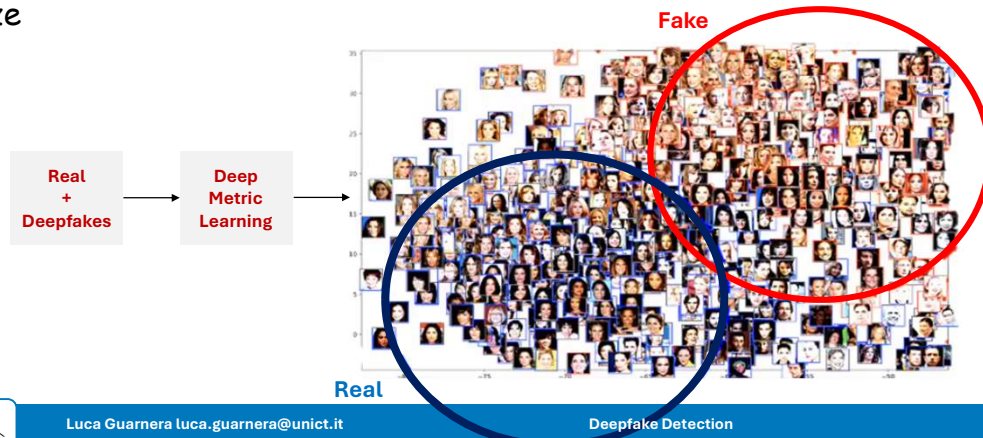
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Potential direction - Anomaly detection

Generalizing to novel Deepfakes is a challenge. Many current SOTA methods fail on unseen Deepfakes of slightly different distributions.

Possible Direction: Deep Metric Learning (e.g. Multi-Similarity Loss) to better generalize



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Deepfakes

Different GANs have different distributions



Different Traces



Deepfake Detection



Deepfake Attribution



Deepfake Model Recognition

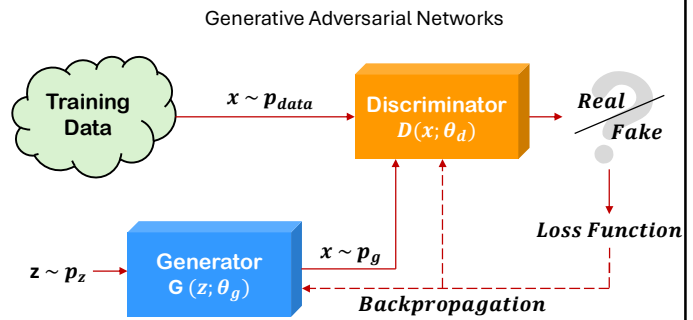
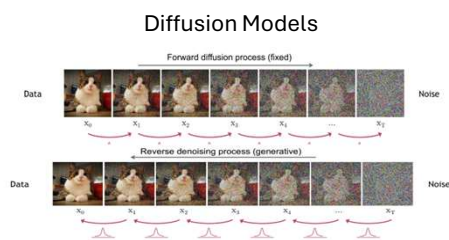
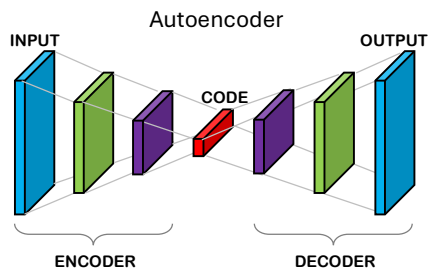


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Autoencoder Vs GAN Vs Diffusion Models

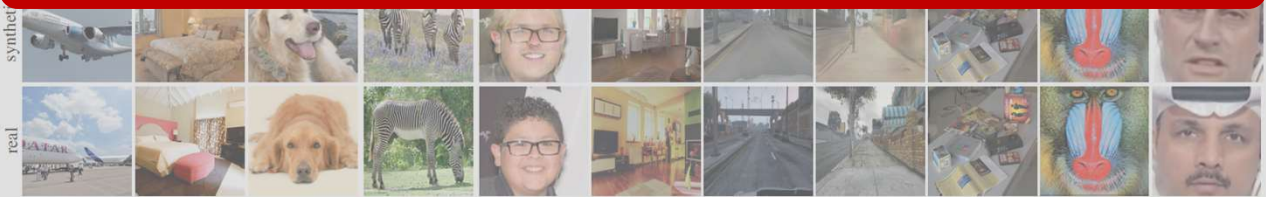


Every technology leaves different traces in the multimedia data it generates

Il problema della semantica



La stessa architettura allenata su un altro dataset con differente semantica rispetto i volti lascerà sui dati generati tracce diverse?



Cosa vedremo nel dettaglio

- Ci focalizzeremo solo sui volti di persone
- Vedremo approcci base, basati su Machine Learning
- Vedremo approcci più avanzati, basati su Deep Learning
- Iniziamo da uno studio preliminare



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Preliminary Analysis

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Deepfake Forensics Analysis

Brief File Format Analysis - WARNING!
Total number of processed images: 1

- Images that are likely to be camera original: 0
- Images with suspicious metadata but no traces of forgery: 0
- Images with traces of possible forgery: 1

Brief File Format Analysis - WARNING!
Total number of processed images: 1

- Images that are likely to be camera original: 0
- Images with suspicious metadata but no traces of forgery: 1
- Images with traces of possible forgery: 0

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Fourier Transform

Example of Analysis carried out with Amped Authenticate software.

(a) Image generated by STARGAN.

(b) Image generated by STYLEGAN.

Each image of both datasets was converted to grayscale (1) and applied progressively: the **Median filter (2)**, the **Laplacian filter (3)**, the **Laplacian filter (4)** applied to the result of (2), the **sum of the results between the Median and Laplacian filters (5)**.

For each operation performed, we show the Fourier transform.

	Grayscale Image (1)	Median (2)	Laplacian (3)	Laplacian (Median) (4)	Laplacian + Median (5)		Grayscale Image (1)	Median (2)	Laplacian (3)	Laplacian (Median) (4)	Laplacian + Median (5)
Image						Image					
							Image				
Image						Image					
							Image				
Image						Image					
							Image				

(a)
(b)

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Deepfake Detector basati su Machine Learning

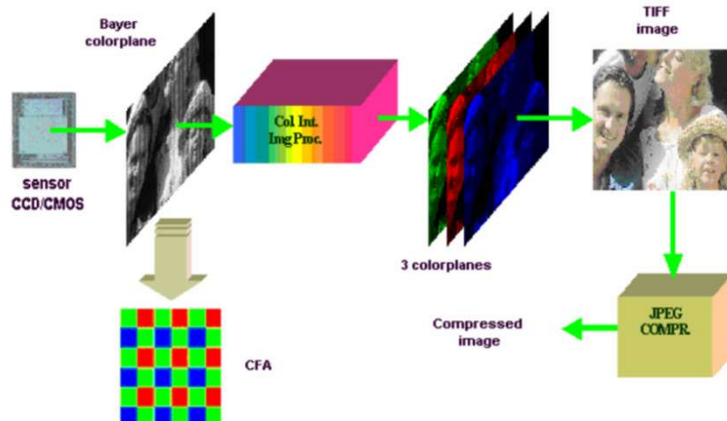
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Facciamo un passo indietro ...

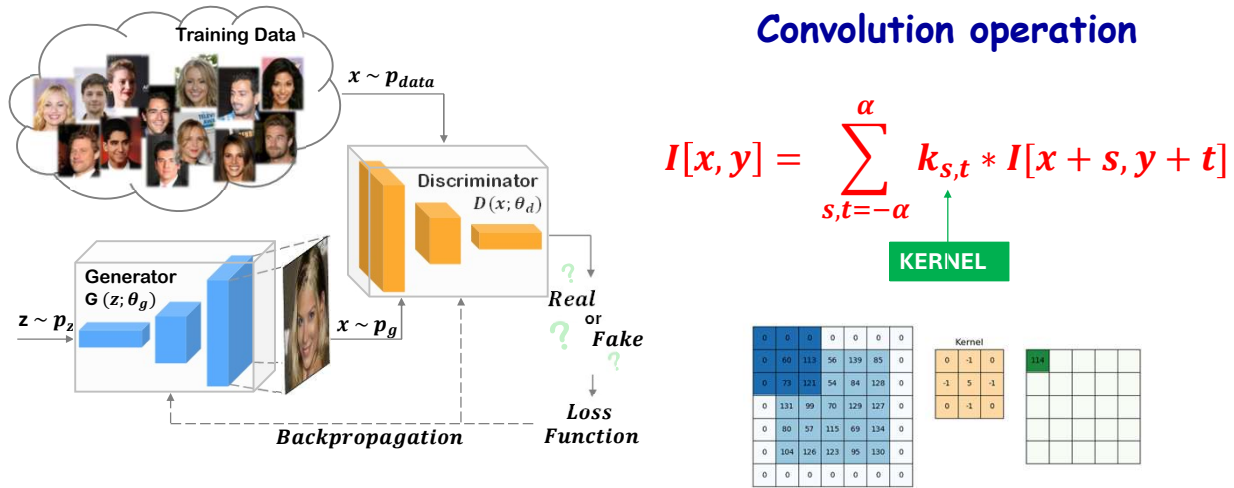


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GAN - Convolution layer

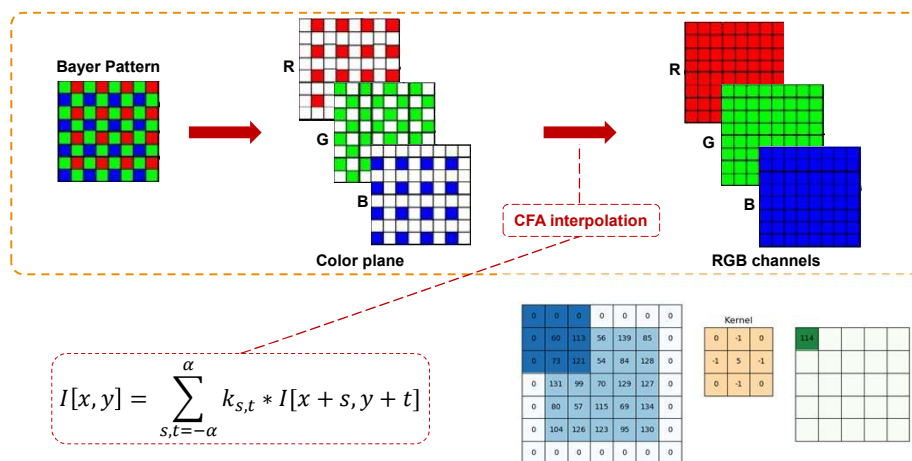


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CFA Interpolation



$$I[x, y] = \sum_{s, t=-\alpha}^{\alpha} k_{s, t} * I[x + s, y + t]$$

Popescu, Alin C., and Hany Farid. "Exposing digital forgeries in color filter array interpolated images." *IEEE Transactions on Signal Processing* 53.10 (2005): 3948-3959.

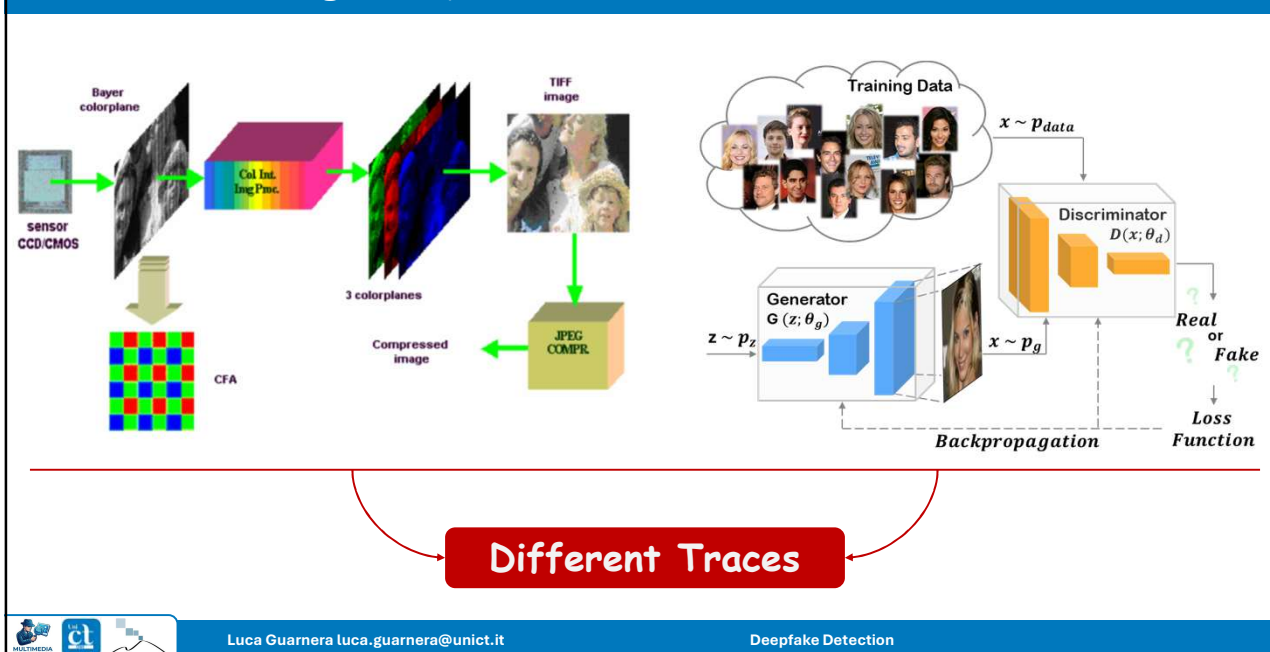


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Image Acquisition Vs Generative Models



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Expectation Maximization Algorithm

L'algoritmo **Expectation-Maximization (EM)** è un metodo iterativo per trovare le stime di massima verosimiglianza dei parametri in modelli statistici con **variabili latenti** (ossia variabili nascoste o non osservate).

Idea di base

Immagina di avere dei dati incompleti: sai *cosa* osservi, ma non *da dove* vengono. L'EM alterna due passi per stimare sia i parametri del modello che l'origine "nascosta" dei dati.

I due passi fondamentali

1. E-step (Expectation)

Dato il valore corrente dei parametri θ , **calcola la probabilità** che ogni dato appartenga a ciascuna componente latente.

"Quanto è probabile che questo punto provenga dal gruppo A, B, o C?"

Formalmente, si calcola la **distribuzione posteriore** delle variabili latenti Z :

$$Q(\theta | \theta_{old}) = E[\log P(X, Z | \theta) | X, \theta_{old}]$$

2. M-step (Maximization)

Usando le probabilità calcolate nell'E-step, **aggiorna i parametri** θ per massimizzare la verosimiglianza attesa.

"Dati questi pesi, quali sono i migliori parametri per ogni gruppo?"

$$\theta_{new} = \operatorname{argmax} Q(\theta | \theta_{old})$$



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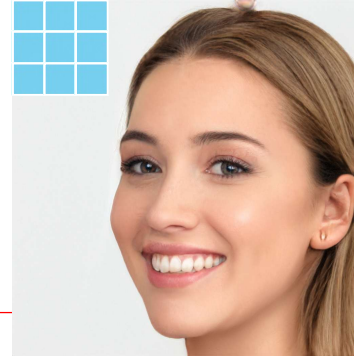
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Expectation Maximization Algorithm

The goal is to **extract the fingerprint** from **input image I** able to numerically represent the **correlations** between each pixel.

$$I[x, y] = \sum_{s, t=-\alpha}^{\alpha} k_{s, t} * I[x + s, y + t]$$

1.6	0.8	1.3
0.1	9	7.1
0.2	3.1	2.4



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Expectation Maximization Algorithm

Assume that the element $I[x, y]$ belongs to one of the following models:

- **M1**: when the element $I[x, y]$ satisfies

$$I[x, y] = \sum_{s, t=-\alpha}^{\alpha} k_{s, t} * I[x + s, y + t]$$

- **M2**: otherwise.

Expectation Maximization (EM) algorithm:

- **Expectation step**: calculates the (density of) probability that each element belongs to a model (M1 or M2);
- **Maximization step**: estimates the (weighted) parameters based on the probabilities of belonging to instances of (M1 or M2).

Pseudo-code of EM Algorithm for Deepfake image Detection

Algorithm 1: Expectation-Maximization Algorithm

Data: Image I
Result: $\hat{k}$
 Initialize N //Kernel size
 Initialize σ_0
 Set $\hat{k}$ random of size $N \times N$
 Set R, P, W matrices with 0 values of the same size as I
 Set p_0 as 1/size of the range of values of I
 for $n = 1; n < 100$ $n++ = 1$ do
 //Expectation Step
 for $\forall$ values in I do
 $R[x, y] = \left| I[x, y] - \sum_{s, t=-\alpha}^{\alpha} k_{s, t} I[x + s, y + t] \right|$
 $P[x, y] = \frac{1}{\sigma_n \sqrt{2\pi}} e^{-\frac{R[x, y]^2}{2\sigma_n^2}}$
 $W[x, y] = \frac{P[x, y]}{P[x, y] + p_0}$
 //Maximization Step
 Calculate $k_{s, t}^{(n+1)}$ as shown in the formula 7



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Expectation Maximization Algorithm

The initial goal is to extract a description, from input **image I**, able to numerically represent the **local correlations** between each pixel in a neighbourhood.

This can be done by means of convolution with a **kernel k** of **NxN size**:

$$I[x, y] = \sum_{s, t=-\alpha}^{\alpha} k_{s, t} * I[x + s, y + t] \quad (1)$$



Expectation Maximization Algorithm

$$I[x, y] = \sum_{s, t=-\alpha}^{\alpha} k_{s, t} * I[x + s, y + t] \Rightarrow$$

The value of the pixel **I[x,y]** is computed considering a **neighborhood** of size **NxN** of the input data.

It is clear that the new estimated information **I[x,y]** mainly **depends on the kernel used in the convolution operation**, which establishes a **mathematical relationship between the pixels**.

For this reason, our goal is to **define a vector k** of size **NxN** able to **capture this hidden and implicit relationship** which characterize of **forensic trace** we want to exploit.



Expectation Maximization Algorithm

Assume that the element $\mathbf{I}[\mathbf{x}, \mathbf{y}]$ belongs to one of the following models:

- **M1**: when the element $\mathbf{I}[\mathbf{x}, \mathbf{y}]$ satisfies

$$I[x, y] = \sum_{s, t=-\alpha}^{\alpha} k_{s, t} * I[x + s, y + t]$$

- **M2**: otherwise.



Expectation Maximization Algorithm

Expectation Maximization (EM) algorithm:

- **Expectation step**: calculates the (density of) probability that each element belongs to a model (M1 or M2);
- **Maximization step**: estimates the (weighted) parameters based on the probabilities of belonging to instances of (M1 or M2).



Expectation Maximization Algorithm

Let's suppose that M_1 and M_2 have different probability distributions with M_1 Gaussian distribution with zero mean and unknown variance and M_2 uniform.

In the **Expectation step**, the Bayes rule that $I[x,y]$ belongs to the model M_1 is computed as follows:

$$\Pr\{I[x,y] \in M_1 \mid I[x,y]\} = \frac{\Pr\{I[x,y] \mid I[x,y] \in M_1\} \Pr\{I[x,y] \in M_1\}}{\sum_{i=1}^2 \Pr\{I[x,y] \mid I[x,y] \in M_i\} \Pr\{I[x,y] \in M_i\}}$$



Expectation Maximization Algorithm

$$\Pr\{I[x,y] \in M_1 \mid I[x,y]\} = \frac{\Pr\{I[x,y] \mid I[x,y] \in M_1\} \Pr\{I[x,y] \in M_1\}}{\sum_{i=1}^2 \Pr\{I[x,y] \mid I[x,y] \in M_i\} \Pr\{I[x,y] \in M_i\}}$$

where the probability distribution of M_1 which represents the probability of observing a sample $I[x,y]$, knowing that it was generated by the model M_1 is:

$$\Pr\{I[x,y] \mid I[x,y] \in M_1\} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{R[x,y]^2}{2\sigma^2}}$$

where

$$R[x,y] = \left| I[x,y] - \sum_{s,t=-\alpha}^{\alpha} k_{s,t} * I[x+s,y+t] \right|$$



Once defined if $\mathbf{I}[x, y]$ belongs to model M_1 (or M_2), the values of the vector $\vec{k}$ are estimated using **least squares method**, minimizing the following:

$$E(\vec{k}) = \sum_{x,y} w[x, y] \left(I[x, y] - \sum_{s,t=-\alpha}^{\alpha} k_{s,t} * I[x + s, y + t] \right)^2 \quad (2)$$

where $w \equiv Pr\{I[x, y] \in M_1 | I[x, y]\}$.

This error function (2) can be minimized by computing the gradient of vector $\vec{k}$. The update of $k_{i,j}$ is carried out by computing the partial derivative of (2) as follows:

$$\frac{\partial E}{\partial k_{i,j}} = 0$$



$$\frac{\partial E}{\partial k_{i,j}} = 0$$

Hence, the following **linear equations system** is obtained:

$$\sum_{s,t=-\alpha}^{\alpha} k_{s,t} \left(\sum_{x,y} w[x, y] I[x + i, y + j] I[x + s, y + t] \right) = \sum_{x,y} w[x, y] I[x + i, y + j] I[x, y]$$

The two steps of the EM algorithm are iteratively repeated.

The **variance** value σ^2 which is still **unknown**, is then **estimated in the Maximization step** (at each i-th iteration).

$$\sigma_i = \sqrt{\frac{\sum_{x,y} w[x, y] R^2[x, y]}{\sum_{x,y} w[x, y]}}$$



Algorithm 1: Expectation-Maximization Algorithm**Data:** Image I **Result:** $\vec{k}$ Initialize N //Kernel sizeInitialize σ_0 Set $\vec{k}$ random of size $N \times N$ Set R, P, W matrices with 0 values of the same size as I Set p_0 as 1/size of the range of values of I **for** $n = 1; n < 100$ $n+1$ **do**

//Expectation Step

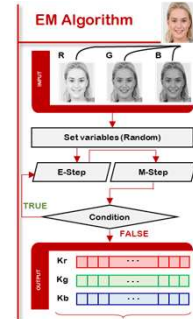
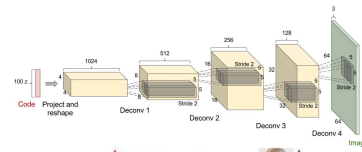
for $\forall$ values in I **do**

$$R[x, y] = \left| I[x, y] - \sum_{s, t=-\alpha}^{\alpha} k_{s, t} I[x + s, y + t] \right|$$

$$P[x, y] = \frac{1}{\sigma_n \sqrt{2\pi}} e^{-\frac{R[x, y]^2}{2\sigma_n^2}}$$

$$W[x, y] = \frac{P[x, y]}{P[x, y] + p_0}$$

//Maximization Step

Calculate $k_{s, t}^{(n+1)}$ as shown in the formula 7

$$\sum_{s, t=-\alpha}^{\alpha} k_{s, t} \left(\sum_{x, y} w[x, y] I[x + i, y + j] I[x + s, y + t] \right) = \sum_{x, y} w[x, y] I[x + i, y + j] I[x, y] \rightarrow k_{s, t}^{n+1} = \frac{\sum_{s, t=-\alpha}^{\alpha} k_{s, t} [\sum_{x, y} W[x, y] I[x + i, y + j] I[x + s, y + t]]}{\sum_{x, y} W[x, y] I[x + i, y + j] I[x, y]}$$

(7)



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Riassumendo**Algorithm 1: Expectation-Maximization Algorithm****Data:** Image I **Result:** $\vec{k}$ Initialize N //Kernel sizeInitialize σ_0 Set $\vec{k}$ random of size $N \times N$ Set R, P, W matrices with 0 values of the same size as I Set p_0 as 1/size of the range of values of I **for** $n = 1; n < 100$ $n+1$ **do**

//Expectation Step

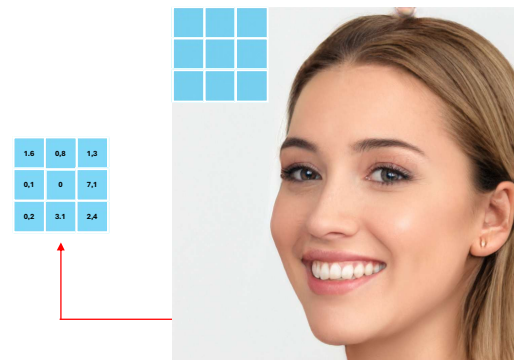
for $\forall$ values in I **do**

$$R[x, y] = \left| I[x, y] - \sum_{s, t=-\alpha}^{\alpha} k_{s, t} I[x + s, y + t] \right|$$

$$P[x, y] = \frac{1}{\sigma_n \sqrt{2\pi}} e^{-\frac{R[x, y]^2}{2\sigma_n^2}}$$

$$W[x, y] = \frac{P[x, y]}{P[x, y] + p_0}$$

//Maximization Step

Calculate $k_{s, t}^{(n+1)}$ as shown in the formula 7

$$k_{s, t}^{n+1} = \frac{\sum_{s, t=-\alpha}^{\alpha} k_{s, t} [\sum_{x, y} W[x, y] I[x + i, y + j] I[x + s, y + t]]}{\sum_{x, y} W[x, y] I[x + i, y + j] I[x, y]}$$



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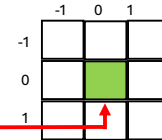
Deepfake Detection

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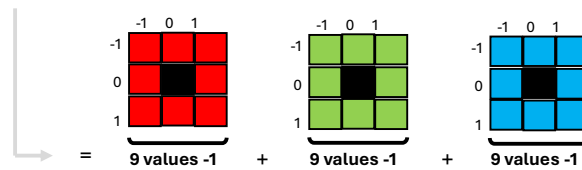
Expectation Maximization Algorithm

The obtained feature vector $\vec{k}$, has dimensions **dependent to parameter α** .

Note that the element $k_{0,0}$ **will always be set equal to 0** ($k_{0,0} = 0$).



Thus, for example, if a **kernel k** with **3x3 size** is employed, the resulting $\vec{k}$ will be a vector of **24 elements** (since the values $k_{0,0}$ are excluded). This is obtained by concatenating the **features extracted from each of the three RGB channels**.

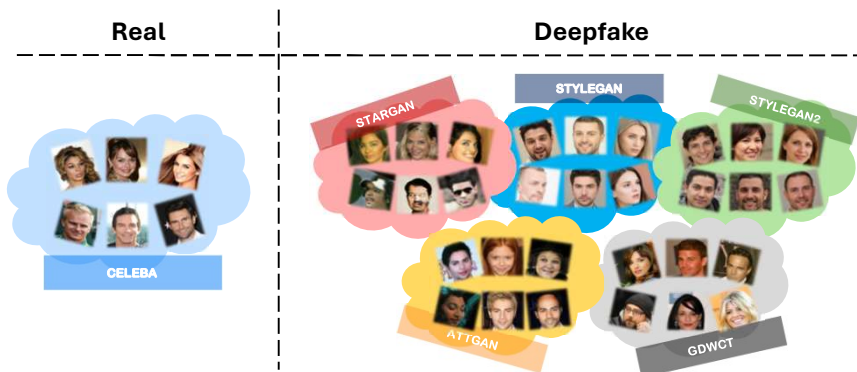


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Deepfake Detection

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Dataset



Guarnera, Luca, Oliver Giudice, and Sebastiano Battiato. "DeepFake Detection by Analyzing Convolutional Traces." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops. 2020.



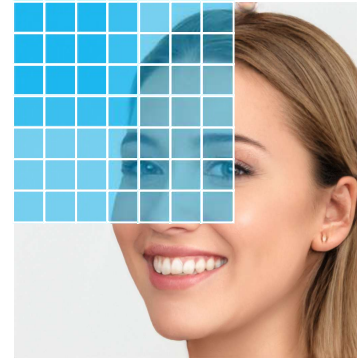
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Deepfake Detection

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Dataset

Method	Number of images generated	Size	Data input to the network	Goal of the network	Kernel size of the latest Convolution Layer
GDWCT	3369	216x216	CELEBA	Improves the stylization capability	4x4
STARGAN	5648	256x256	CELEBA	Image-to-image translations on multiple domains using a single model	7x7
ATTGAN	6005	256x256	CELEBA	Transfer of face attributes with classification constraints	4x4
STYLEGAN	9999	1024x1024	CELEBA-HQ FFHQ	Transfer semantic content from a source domain to a target domain characterized by a different style	3x3
STYLEGAN2	3000	1024x1024	FFHQ	Transfer semantic content from a source domain to a target domain characterized by a different style	3x3



Guarnera, Luca, Oliver Giudice, and Sebastiano Battiato. "DeepFake Detection by Analyzing Convolutional Traces." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops. 2020.

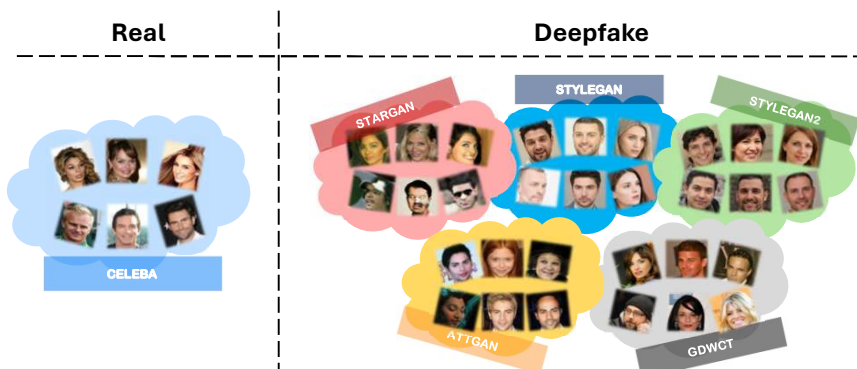


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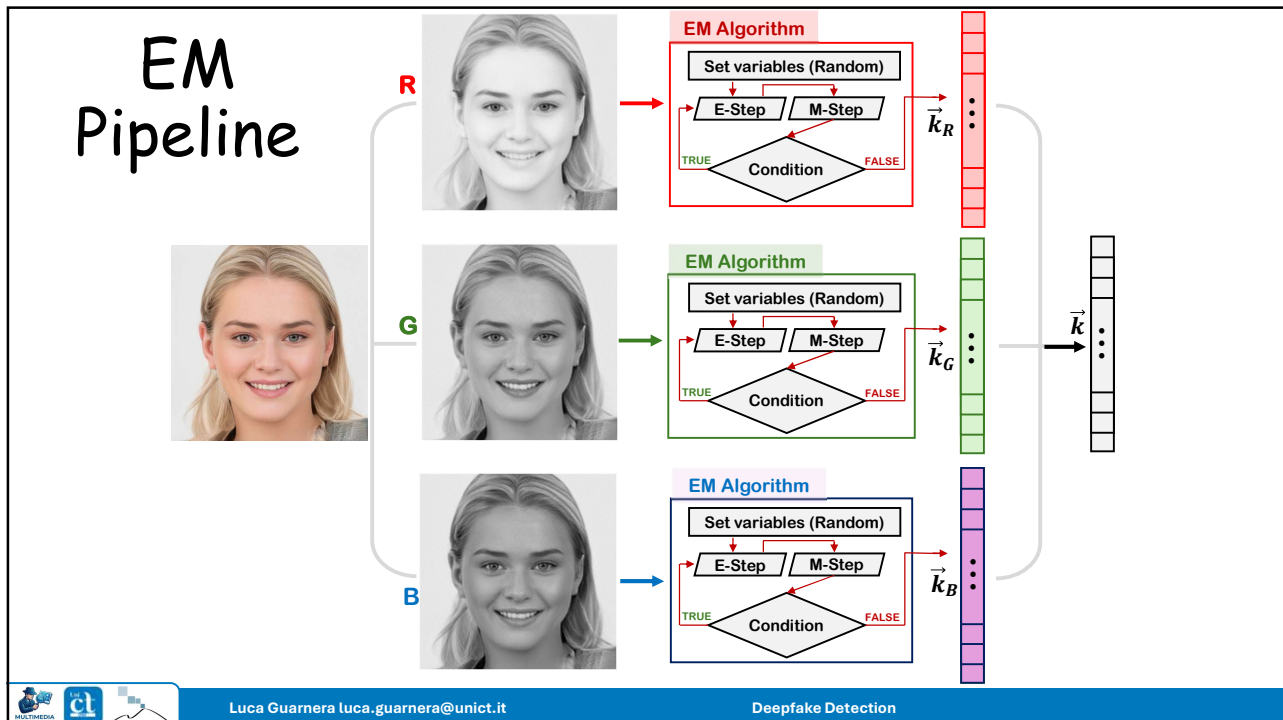
Dataset



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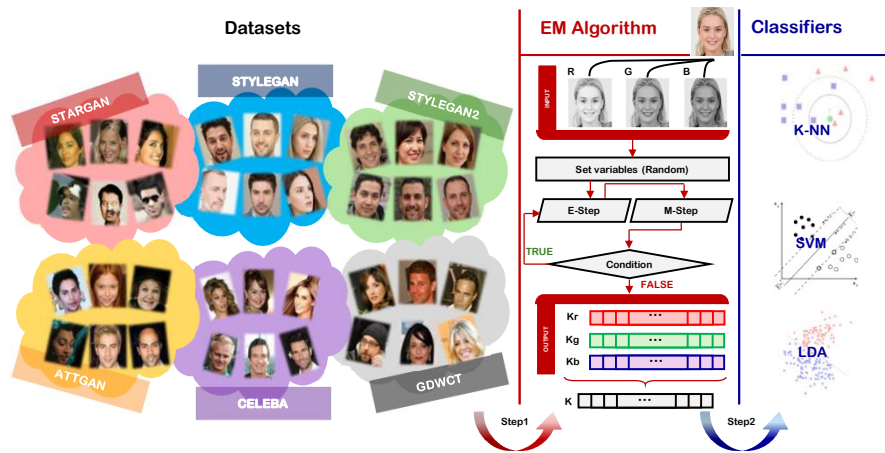
Dataset

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GDWCT	3369	216x216	CELEBA	Improves the stylization capability	4x4
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STYLEGAN2	3000	1024x1024	FFHQ	Transfer semantic content from a source domain to a target domain characterized by a different style	3x3

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Pipeline



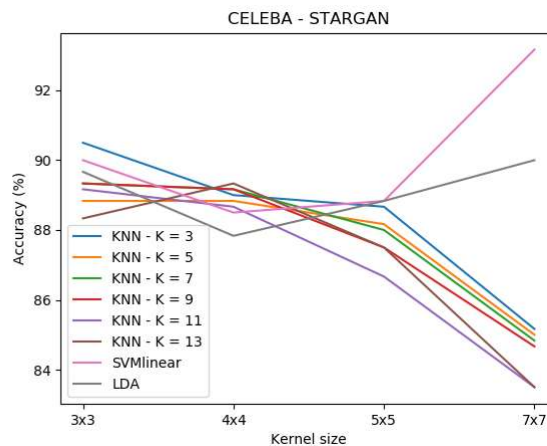
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Deepfake Detection

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CELEBA Vs STARGAN

	Kernel 3x3	Kernel 4x4	Kernel 5x5	Kernel 7x7
KNN, k = 3	90.50	89.00	88.67	85.17
KNN, k = 5	88.83	88.83	88.17	85.00
KNN, k = 7	89.33	89.17	88.00	84.83
KNN, k = 9	89.33	89.17	87.50	84.67
KNN, k = 11	89.17	88.67	86.67	83.50
KNN, k = 13	88.33	89.33	87.50	83.50
SVMLinear	90.00	88.50	88.83	93.17
LDA	89.67	87.83	88.83	90.00



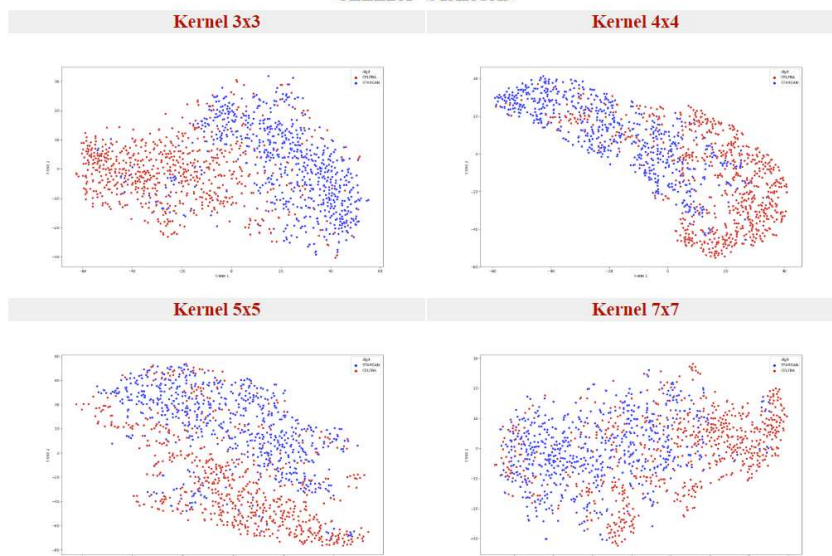
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Deepfake Detection

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t-SEN two dimensions

CELEBA - STARGAN



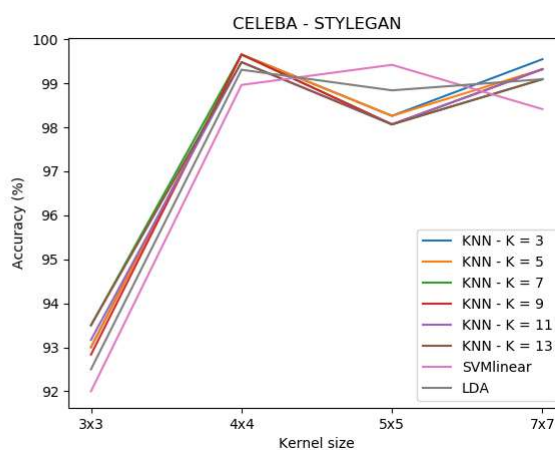
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CELEBA - STYLEGAN

	Kernel 3x3	Kernel 4x4	Kernel 5x5	Kernel 7x7
KNN, k = 3	93.00	99.65	98.26	99.55
KNN, k = 5	93.00	99.65	98.26	99.32
KNN, k = 7	93.50	99.65	98.07	99.09
KNN, k = 9	92.83	99.65	98.07	99.32
KNN, k = 11	93.17	99.48	98.07	99.32
KNN, k = 13	93.50	99.48	98.07	99.09
SVMLinear	92.00	98.96	99.42	98.41
LDA	92.50	99.31	98.84	99.09



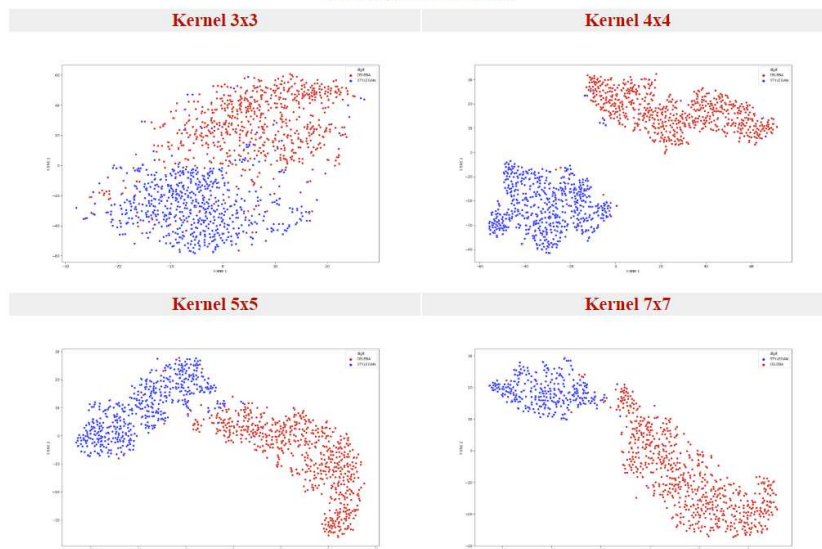
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Deepfake Detection

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t-SEN two dimensions

CELEBA - STYLEGAN



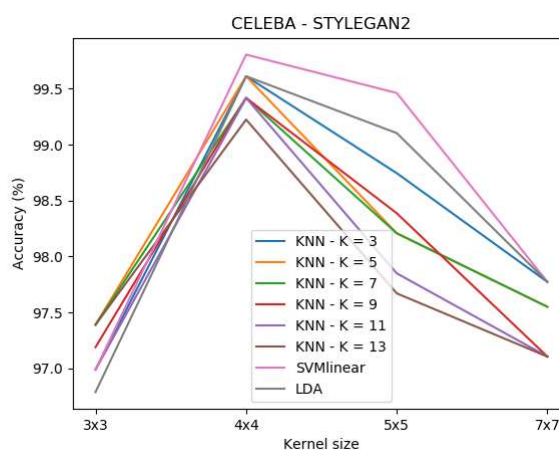
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CELEBA - STYLEGAN2

	Kernel 3x3	Kernel 4x4	Kernel 5x5	Kernel 7x7
KNN, k = 3	96.99	99.61	98.75	97.77
KNN, k = 5	97.39	99.61	98.21	97.55
KNN, k = 7	97.39	99.42	98.21	97.55
KNN, k = 9	97.19	99.42	98.39	97.10
KNN, k = 11	96.99	99.42	97.85	97.10
KNN, k = 13	97.39	99.22	97.67	97.10
SVMLinear	96.99	99.81	99.46	97.77
LDA	96.79	99.61	99.10	97.77



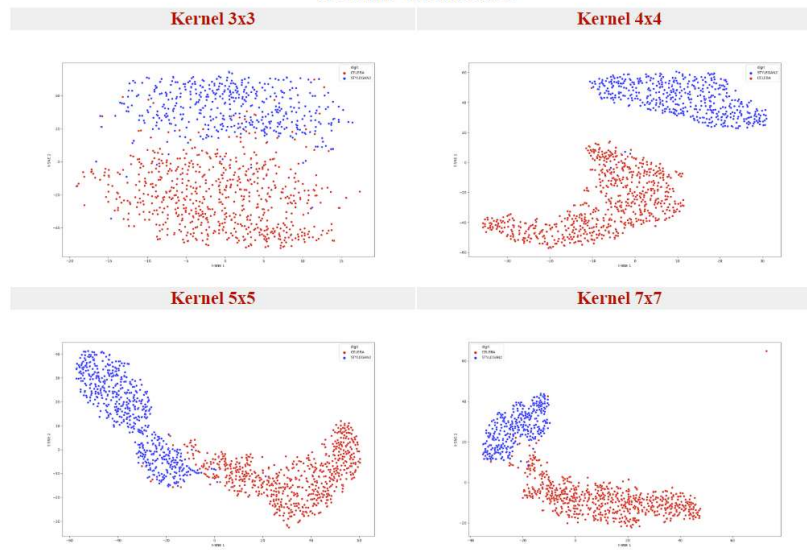
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t-SEN two dimensions

CELEBA - STYLEGAN2



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StyleGAN Vs StyleGAN2

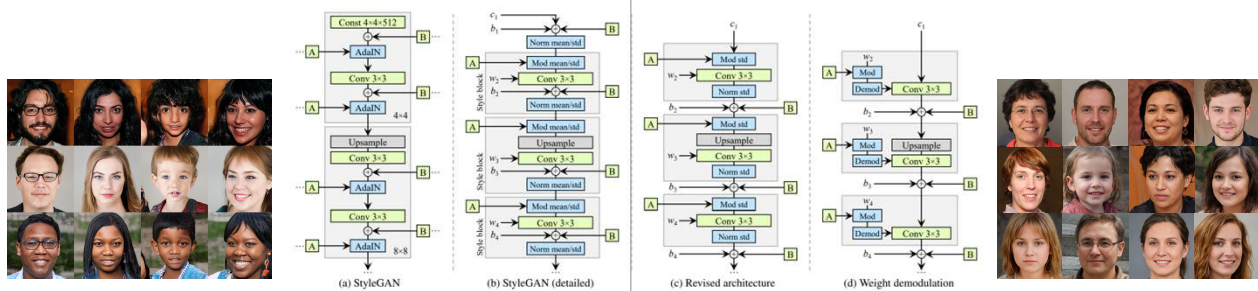


Figure 2. We redesign the architecture of the StyleGAN synthesis network. (a) The original StyleGAN, where $\mathbf{A}$ denotes a learned affine transform from $\mathcal{W}$ that produces a style and $\mathbf{B}$ is a noise broadcast operation. (b) The same diagram with full detail. Here we have broken the AdaIN to explicit normalization followed by modulation, both operating on the mean and standard deviation per feature map. We have also annotated the learned weights (w), biases (b), and constant input (c), and redrawn the gray boxes so that one style is active per box. The activation function (leaky ReLU) is always applied right after adding the bias. (c) We make several changes to the original architecture that are justified in the main text. We remove some redundant operations at the beginning, move the addition of b and $\mathbf{B}$ to be outside active area of a style, and adjust only the standard deviation per feature map. (d) The revised architecture enables us to replace instance normalization with a "demodulation" operation, which we apply to the weights associated with each convolution layer.

Karras, Tero, et al. "Analyzing and improving the image quality of stylegan." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2020.



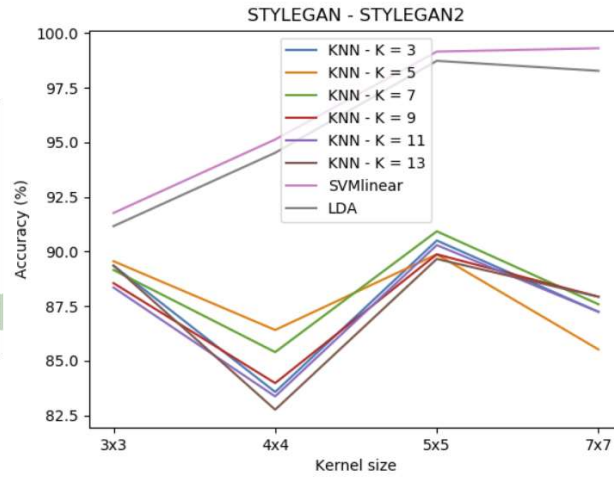
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StyleGAN Vs StyleGAN2

	Kernel 3x3	Kernel 4x4	Kernel 5x5	Kernel 7x7
KNN, k = 3	89.36	83.57	90.51	87.24
KNN, k = 5	89.56	86.41	89.87	85.52
KNN, k = 7	89.16	85.40	90.93	87.59
KNN, k = 9	88.55	83.98	89.87	87.93
KNN, k = 11	88.35	83.37	90.30	87.24
KNN, k = 13	89.36	82.76	89.66	87.93
SVMLinear	91.77	95.13	99.16	99.31
LDA	91.16	94.52	98.73	98.28



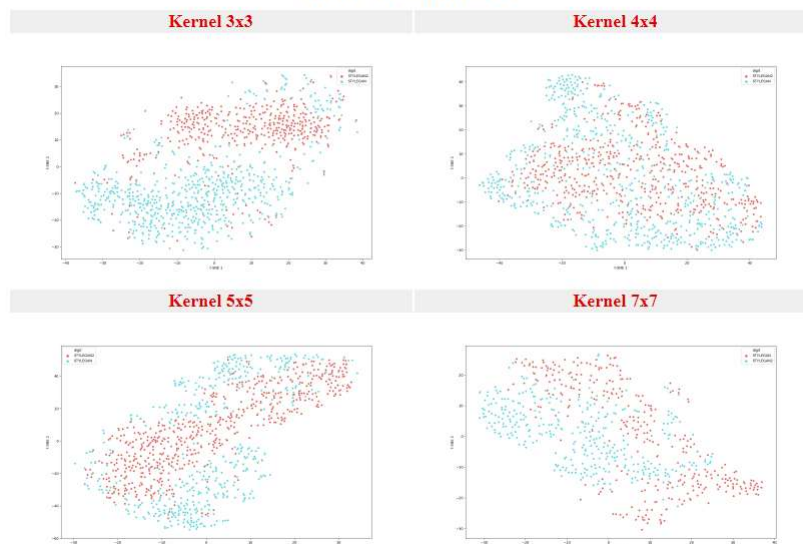
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Deepfake Detection

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t-SEN two dimensions

STYLEGAN - STYLEGAN2



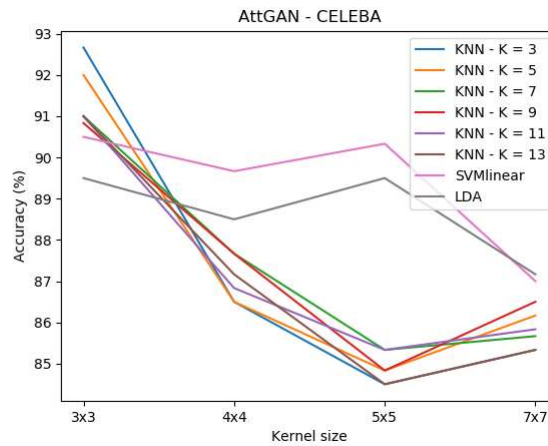
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Deepfake Detection

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CELEBA Vs AttGAN

	Kernel 3x3	Kernel 4x4	Kernel 5x5	Kernel 7x7
KNN, k = 3	92.67	86.50	84.50	85.33
KNN, k = 5	92.00	86.50	84.83	86.17
KNN, k = 7	91.00	87.67	85.33	85.67
KNN, k = 9	90.83	87.67	84.83	86.50
KNN, k = 11	91.00	86.83	85.33	85.83
KNN, k = 13	91.00	87.17	84.50	85.33
SVMLinear	90.50	89.67	90.33	87.00
LDA	89.50	88.50	89.50	87.17



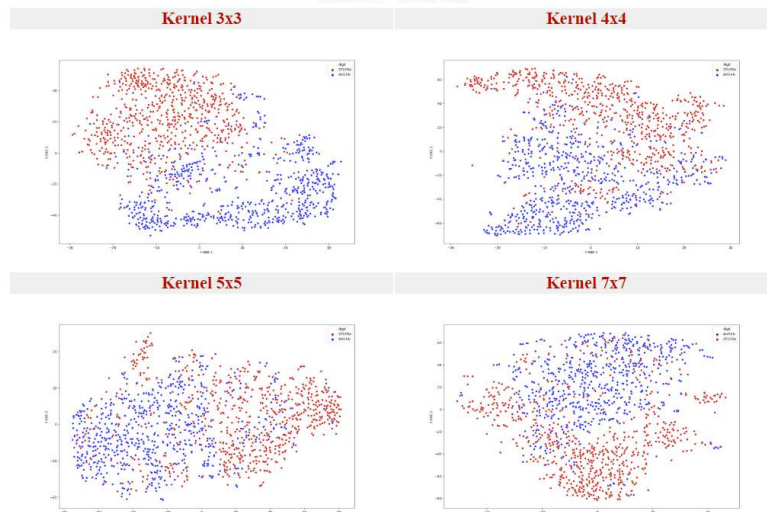
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Deepfake Detection

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t-SEN two dimensions

AttGAN - CELEBA



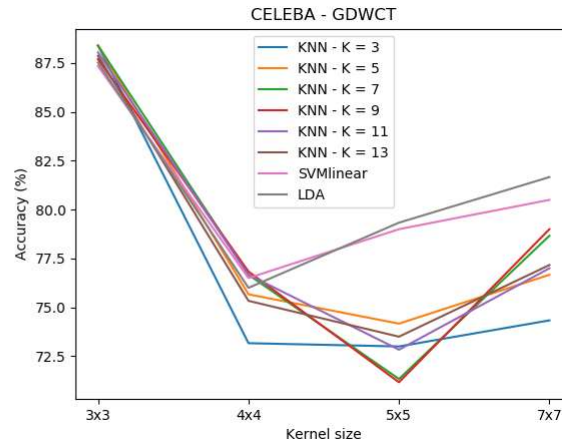
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Deepfake Detection

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CELEBA Vs GDWCT

	Kernel 3x3	Kernel 4x4	Kernel 5x5	Kernel 7x7
KNN, k = 3	88.40	73.17	73.00	74.33
KNN, k = 5	88.40	75.67	74.17	76.67
KNN, k = 7	88.40	76.67	71.33	78.67
KNN, k = 9	87.70	76.83	71.17	79.00
KNN, k = 11	88.05	76.67	72.83	77.00
KNN, k = 13	87.87	75.33	73.50	77.17
SVMLinear	87.35	76.50	79.00	80.50
LDA	87.52	76.00	79.33	81.67



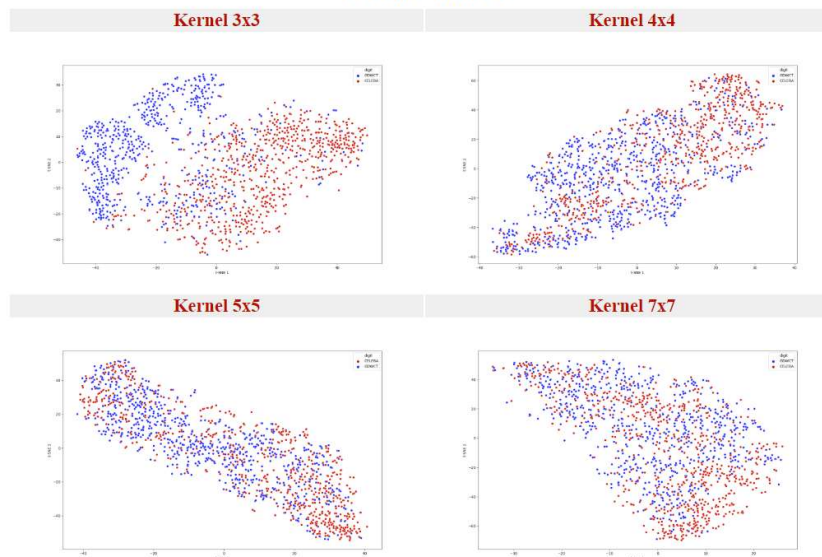
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Deepfake Detection

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t-SEN two dimensions

CELEBA - GDWCT



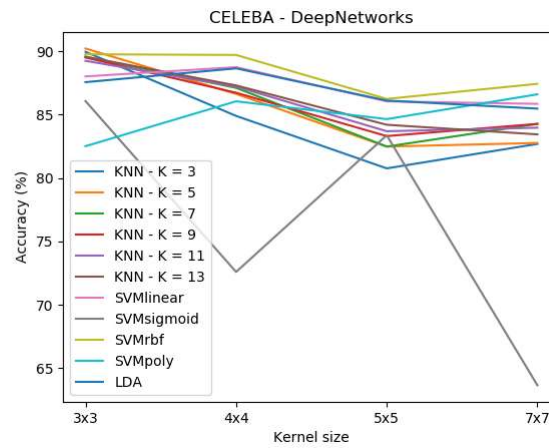
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Deepfake Detection

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CELEBA - DeepNetworks

	Kernel 3x3	Kernel 4x4	Kernel 5x5	Kernel 7x7
KNN, k = 3	89.96	84.90	80.76	82.69
KNN, k = 5	90.22	86.63	82.48	82.77
KNN, k = 7	89.57	87.12	82.48	84.27
KNN, k = 9	89.51	86.73	83.31	84.27
KNN, k = 11	89.25	87.21	83.69	83.97
KNN, k = 13	89.57	87.31	84.20	83.45
SVMLinear	88.02	88.75	86.05	85.85
SVMsigmoid	86.08	72.60	83.38	83.66
SVMrbf	89.77	89.71	86.24	87.43
SVMPoly	82.51	86.06	84.65	86.61
LDA	87.56	88.65	86.11	85.48



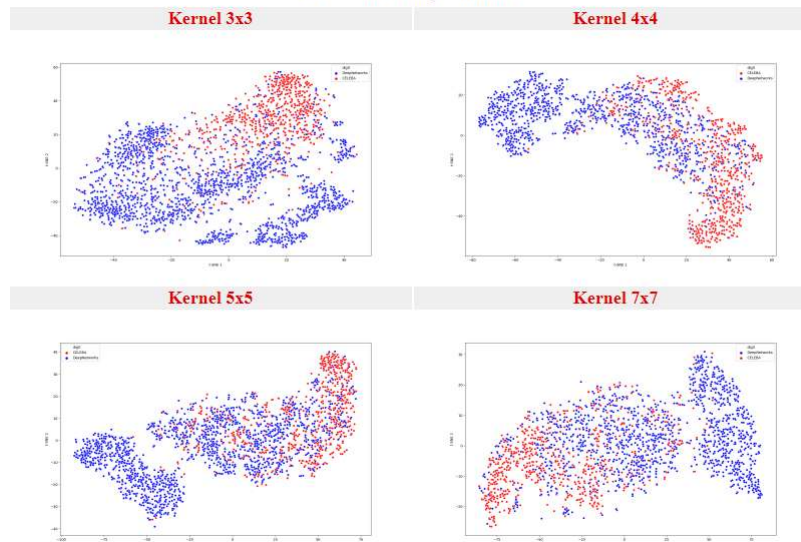
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t-SEN two dimensions

CELEBA - DeepNetworks



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Deepfake Detection

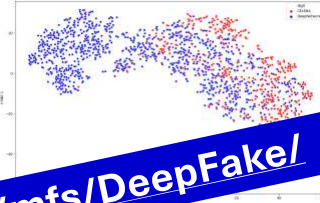
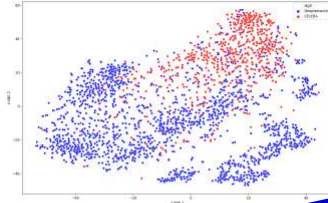
74

t-SEN two dimensions

CELEBA - DeepNetworks

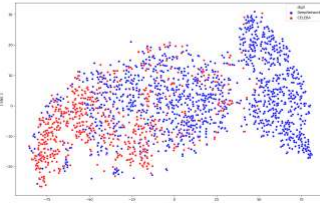
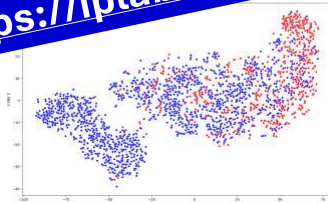
Kernel 3x3

Kernel 4x4



Kernel 5x5

Kernel 7x7



<https://iplab.dmi.unict.it/mfs/DeepFake/>



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Robustness Experiments



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Robustness Experiments

	ATTGAN			GDWCT			STARGAN			STYLEGAN			STYLEGAN2		
	Kernel Size			Kernel Size			Kernel Size			Kernel Size			Kernel Size		
	3x3	5x5	7x7	3x3	5x5	7x7	3x3	5x5	7x7	3x3	5x5	7x7	3x3	5x5	7x7
Raw Images	92.99	88.51	87.11	91.58	77.41	82.01	88.54	84.43	90.55	95.29	99.48	99.30	96.91	99.64	99.32
Random Square	82.54	75.47	75	62.03	61.54	63.27	81.16	78.95	76.19	97.26	100	97.37	99.02	100	100
Gaussian Blur, kernel size = 3x3	77.78	73.58	72.22	56.96	59.38	61.22	73.91	80.7	61.9	93.15	98.33	92.11	96.08	98.81	96.08
Gaussian Blur, kernel size = 9x9	76.19	76.92	68.57	56.96	67.19	61.22	72.46	77.19	64.29	97.26	100	94.59	96.08	97.62	94.12
Gaussian Blur, kernel size = 15x15	80.95	76.92	77.14	64.56	67.69	57.14	82.61	80.7	75.61	97.26	98.33	94.59	100	97.59	98.04
Rotation 45°	90	84.31	85.29	67.53	73.02	66.67	85.29	82.14	87.8	89.04	91.67	91.89	97.4	94.2	97.62
Rotation 90°	100	94.23	100	93.59	92.19	93.75	92.75	92.98	97.56	100	100	97.3	100	100	100
Rotation 180°	83.87	86.54	82.86	74.36	67.19	59.18	84.06	91.23	78.57	100	100	91.89	97.03	98.8	98.04
Scaling +50%	88.71	78.43	91.18	78.21	71.88	68.09	89.71	83.93	90	97.22	100	97.3	99	98.78	100
Scaling -50%	75.81	78.85	77.78	71.79	57.81	68.09	79.71	64.91	64.29	95.83	96.67	100	99.01	97.59	94.23
JPEG Compression	86.69	91.67	91.18	85.17	89.33	84.66	89.17	92.69	92.01	99.5	99.33	97.57	99.49	98.96	98.55

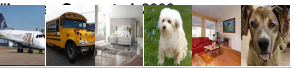
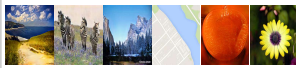


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Deepfake Detection

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Real Vs Deepfake

STARGAN Image-to-image translations on multiple domains using one model Input: CELEBA Output image size: 256x256 #Images Generated: 2000 	ATTGAN Transfers face attributes with constraints Input: CELEBA Output image size: 256x256 #Images Generated: 2000 	STYLEGAN2 Improves STYLEGAN quality with the same task Input: FFHQ Output image size: 1024x1024 #Images Generated: 2000 	PROGAN Creates images starting from low quality details Input: LSUN Output image size: 256x256 
GDWCT Improves the styling capability Input: CELEBA Output image size: 216x216 #Images Generated: 2000 	IMLE Synthesizes images given a semantic layout Input: GTA Output image size: 512x216 #Images Generated: 2000 	CYCLEGAN Image-to-image translation for everything Input: Cityscapes, CMP Facade, Google Maps, Zappos50K, ImageNet, Flickr API. Output image size: 256x256 #Images Generated: 2000 	SPADE Synthesizes photorealistic images from a semantic layout Input: ADE20K Output image size: 256x256 #Images Generated: 2000 
STYLEGAN Transfers semantic content from a source domain to a target domain characterized by a different style Input: FFHQ Output image size: 1024x1024 #Images Generated: 2000 	FACE-FORENSICS++ It is not an architecture but a dataset of manipulated videos with four methods Input: Youtube Videos Output image size: min: 162x162 max: 895x895 #Images Generated: 2000 		

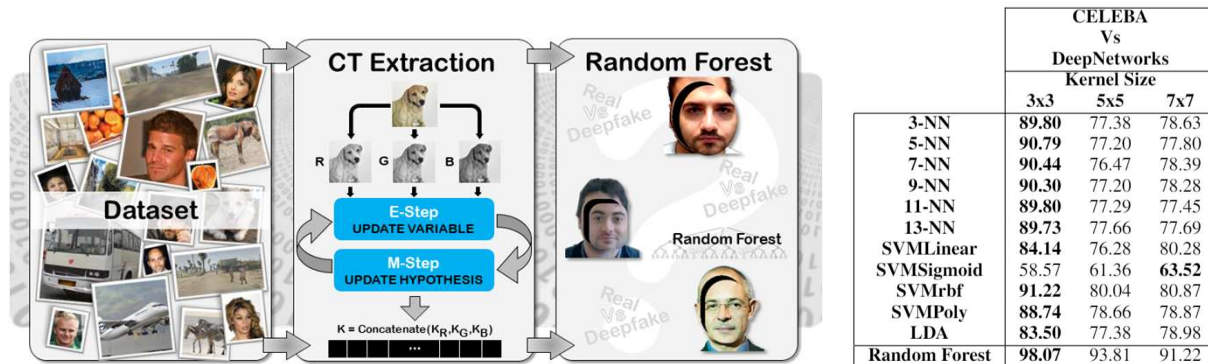


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Deepfake Detection

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Real Vs Deepfake (Binary classification)

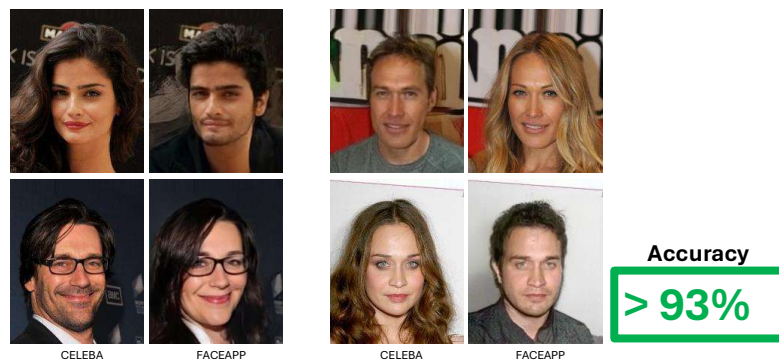


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Test with FaceApp



Guarnera, Luca, Oliver Giudice, and Sebastiano Battiato. "Fighting Deepfake by Exposing the Convolutional Traces on Images." *IEEE Access* 8 (2020): 165085-165098.



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Deepfake Detection

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Fighting deepfakes by detecting GAN DCT anomalies

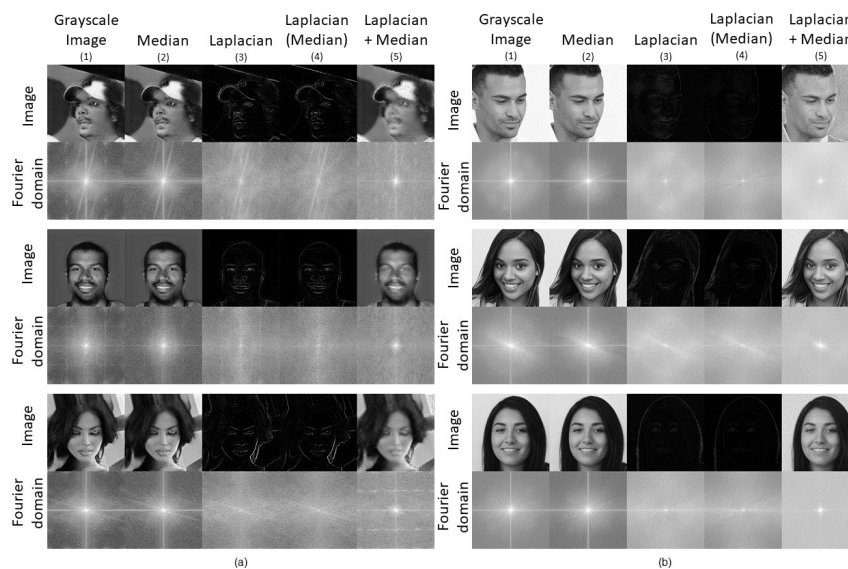
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Deepfake Detection



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Fourier Transform

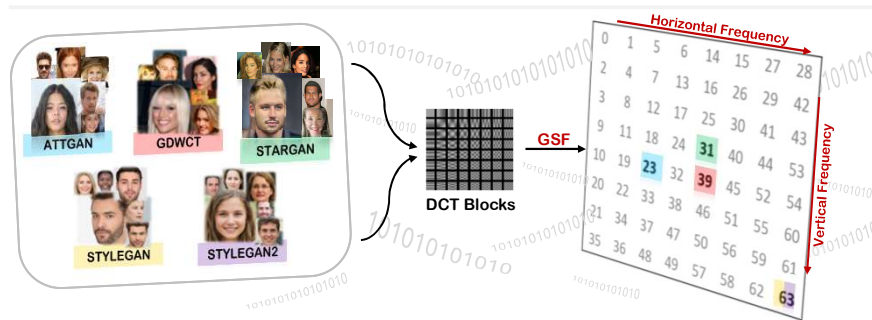


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CTF-DCT approach



CTF-DCT approach

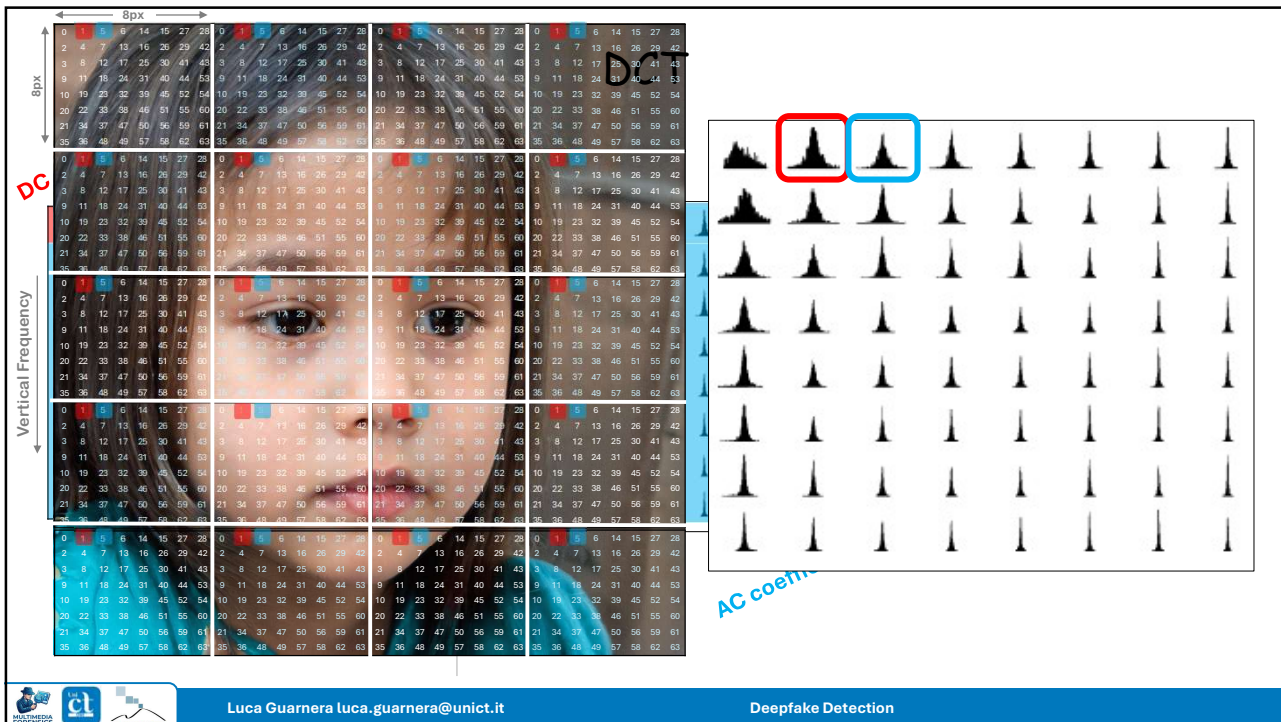
- Let $\mathbf{I}$ be a digital image.
- Following the **JPEG pipeline**, $\mathbf{I}$ is divided into non-overlapping blocks of size **8x8**.
- The **Discrete Cosine Transform (DCT)** is then applied to each block, formally:

$$F[u, v] = \frac{1}{4} C(u) C(v) \left[\sum_{x=0}^7 \sum_{y=0}^7 I[x, y] \cos(a) \cos(b) \right]$$

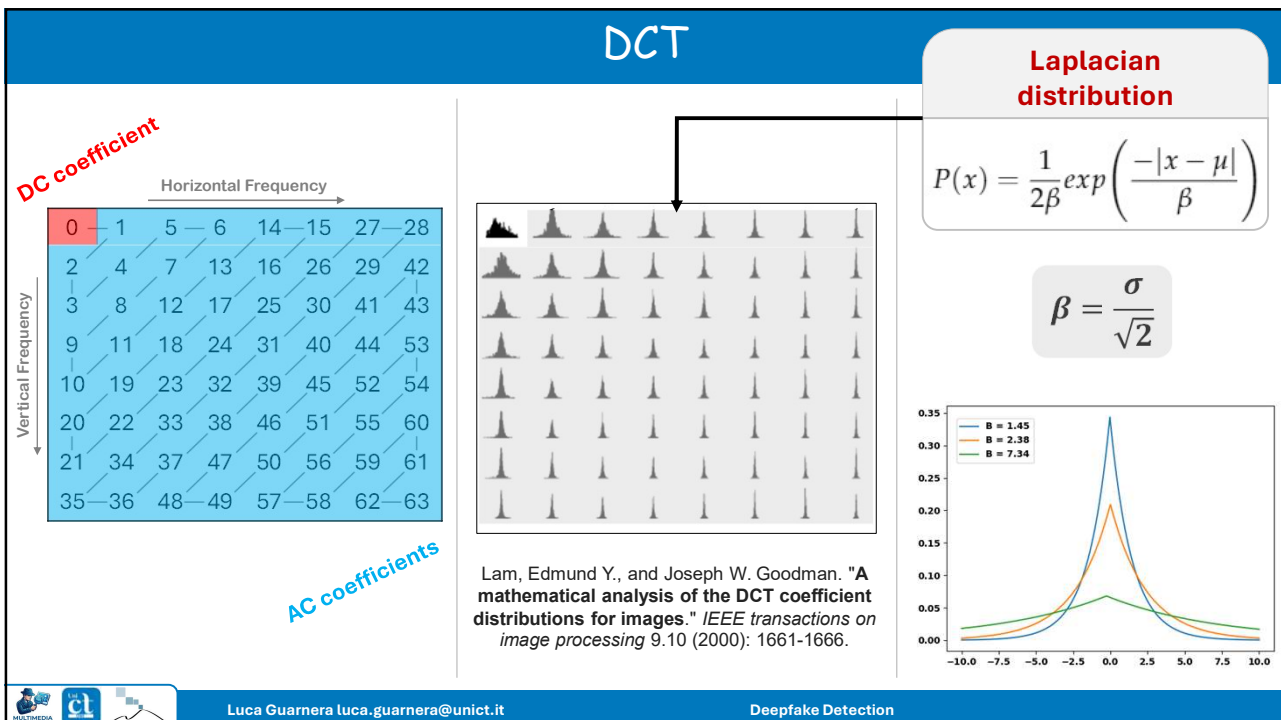
where

$$a = \frac{(2x+1)u\pi}{16}, b = \frac{(2y+1)v\pi}{16}, C(u) = \begin{cases} \frac{1}{\sqrt{2}} & u = 0 \\ 1 & u > 0 \end{cases}, C(v) = \begin{cases} \frac{1}{\sqrt{2}} & v = 0 \\ 1 & v > 0 \end{cases}$$

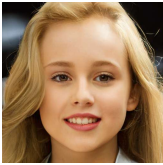




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$\vec{\beta}_I = \{\beta_{I_1}, \beta_{I_2}, \dots, \beta_{I_N}\}$
 with $N = 63$ (DC coefficient is excluded) the corresponding **feature vector of the image I**

Image I

We exploited related statistics on different image-datasets $\mathbf{DT}_g$ with

$g = \{\text{StarGAN, AttGAN, GDWCT, StyleGAN, StyleGAN2, CelebA, FFHQ}\}$
 E.g. $\mathbf{DT}_{\text{StyleGAN}}$ represents all the available images generated by StyleGAN engine.

For **each image-set** $\mathbf{DT}_g$ let's consider the following representation:

$$\beta_{\mathbf{DT}_g} = \begin{pmatrix} \vec{\beta}_1 \\ \vec{\beta}_2 \\ \vdots \\ \vec{\beta}_{|\mathbf{DT}_g|} \end{pmatrix} = \begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \dots & \beta_{1,63} \\ \beta_{2,1} & \beta_{2,2} & \dots & \beta_{2,63} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{|\mathbf{DT}_g|,1} & \beta_{|\mathbf{DT}_g|,2} & \dots & \beta_{|\mathbf{DT}_g|,63} \end{pmatrix}$$

where $|\mathbf{DT}_g|$ is the number of images in $\mathbf{DT}_g$ ($|\mathbf{DT}_g| = k, k = 3.000$).

$\mathbf{DT}_{\text{StarGAN}}$



$\text{Image}_1 \rightarrow \vec{\beta}_1$



$\text{Image}_2 \rightarrow \vec{\beta}_2$

$\vdots$



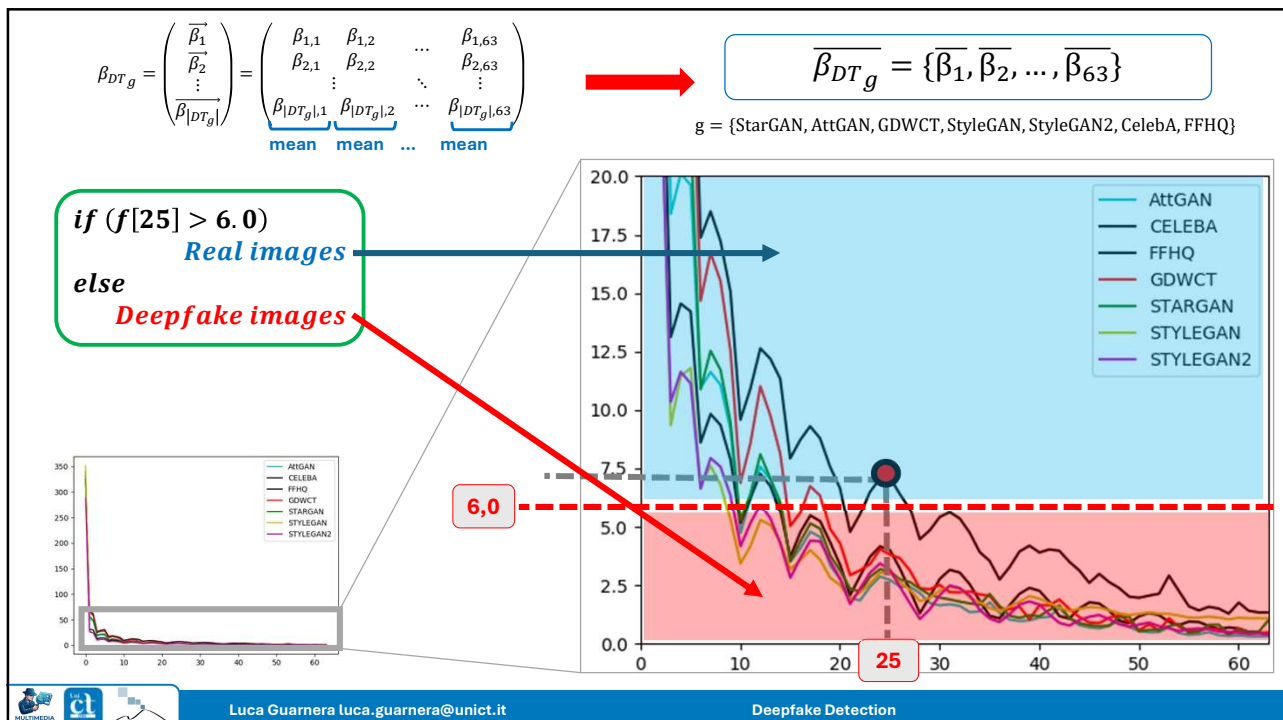
$\text{Image}_{|\mathbf{DT}_{\text{StarGAN}}|} \rightarrow \vec{\beta}_{|\mathbf{DT}_{\text{StarGAN}}|}$



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GAN Specific Frequencies

We computed a χ^2 **distance** as follows:

$$(1) \quad \chi^2(\beta_{DT_i}, \beta_{DT_j}) = \sum_{r=1}^k \frac{(\beta_{DT_i}[r, c] - \beta_{DT_j}[r, c])^2}{\beta_{DT_i}[r, c]} \quad \text{with } c = 1, \dots, 63$$

$$\beta_{DT_g} = \begin{pmatrix} \vec{\beta_1} \\ \vec{\beta_2} \\ \vdots \\ \vec{\beta_{|DT_g|}} \end{pmatrix} = \begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \dots & \beta_{1,63} \\ \beta_{2,1} & \beta_{2,2} & \dots & \beta_{2,63} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{|DT_g|,1} & \beta_{|DT_g|,2} & \dots & \beta_{|DT_g|,63} \end{pmatrix}$$

where $i, j \in g$, $i \neq j$, c is the column which corresponds to the AC coefficient and r are the rows in β_{DT_g} that represents all β_i features. The distance $\chi^2(\beta_{DT_i}, \beta_{DT_j})$ is a vector with size equal to 63.

Finally, it is possible to define the **GAN Specific Frequency (GSF)** as follows:

$$GSF_{DT_i, DT_j} = \underset{c}{\operatorname{argmax}} \chi^2(\beta_{DT_i}, \beta_{DT_j}) = \underset{c}{\operatorname{argmax}} \sum_{r=1}^k \frac{(\beta_{DT_i}[r, c] - \beta_{DT_j}[r, c])^2}{\beta_{DT_i}[r, c]} \quad \text{with } c = 1, \dots, 63 \quad (2)$$

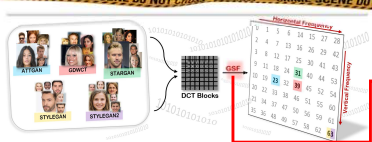


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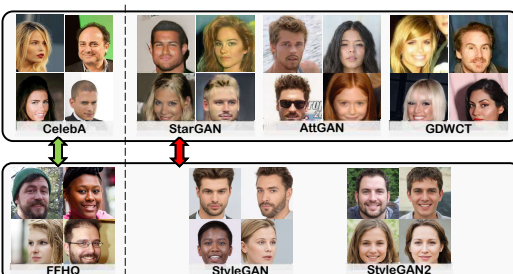
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CTF-DCT approach



$$\beta_{DT_g} = \begin{pmatrix} \vec{\beta_1} \\ \vec{\beta_2} \\ \vdots \\ \vec{\beta_{|DT_g|}} \end{pmatrix} = \begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \dots & \beta_{1,63} \\ \beta_{2,1} & \beta_{2,2} & \dots & \beta_{2,63} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{|DT_g|,1} & \beta_{|DT_g|,2} & \dots & \beta_{|DT_g|,63} \end{pmatrix}$$

$$GSF_{DT_i, DT_j} = \underset{c}{\operatorname{argmax}} \chi^2(\beta_{DT_i}, \beta_{DT_j}) = \underset{c}{\operatorname{argmax}} \sum_{r=1}^k \frac{(\beta_{DT_i}[r, c] - \beta_{DT_j}[r, c])^2}{\beta_{DT_i}[r, c]} \quad \text{with } c = 1, \dots, 63$$



Real

Deepfake

	StarGAN		AttGAN		GDWCT		StyleGAN		StyleGAN2	
	GSF	Acc(%)	GSF	Acc(%)	GSF	Acc(%)	GSF	Acc(%)	GSF	Acc(%)
CelebA	31	94%	23	89%	39	91%	2	100%	2	100%
FFHQ	63	96%	2	95%	2	94%	63	99%	63	99%
StarGAN			63	100%	63	95%	50	100%	63	100%
AttGAN					7	72%	62	99%	2	95%
GDWCT							2	98%	2	99%
StyleGAN									54	99%

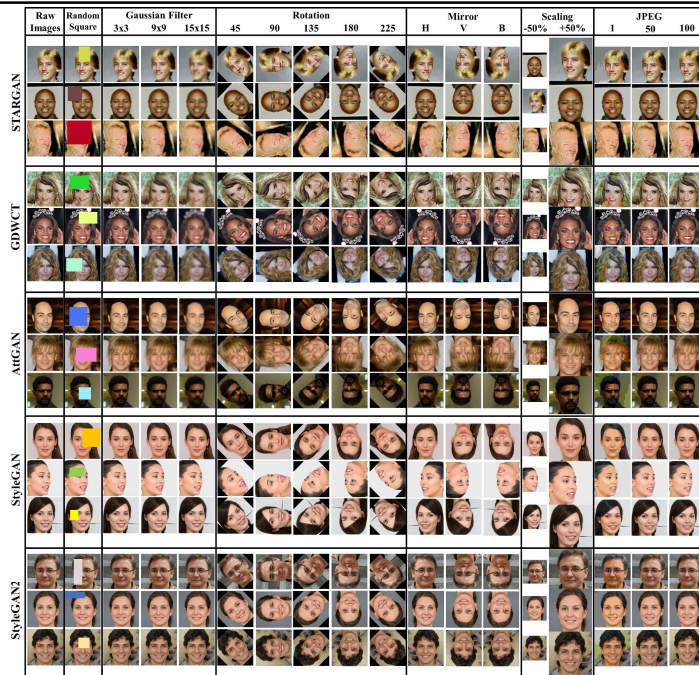


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Test with Noise



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Test with Noise

		Real			AtGAN			GDWCT			StarGAN			StyleGAN			StyleGAN2			Overall Accuracy
		Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1	
Raw Images		99	97	98	99	100	99	98	98	98	99	100	100	99	98	99	98	100	99	99
Random Square		98	94	96	90	96	93	92	89	91	100	98	99	98	99	98	99	99	99	96
Gaussian Filter	3x3	98	95	96	83	88	86	89	92	91	92	86	89	97	98	98	99	99	99	93
	9x9	98	99	98	62	59	60	70	79	74	59	53	56	99	98	99	98	99	98	81
	15x15	100	97	98	58	64	61	72	64	68	55	53	54	98	99	98	95	100	97	80
Rotation	45°	97	93	95	85	82	83	92	98	95	84	84	84	97	99	98	99	98	98	92
	90°	98	99	98	95	99	97	98	93	95	100	99	99	99	98	98	99	99	99	98
	135°	95	96	96	85	83	84	97	94	96	83	86	85	96	95	96	96	97	97	92
	180°	98	94	96	95	100	97	97	95	96	99	100	99	98	99	99	100	99	99	98
	225°	96	95	95	88	85	87	96	96	96	86	89	88	96	97	97	97	97	97	93
Mirror	H	99	96	98	99	100	99	98	99	98	99	100	99	99	99	99	100	100	100	99
	V	99	96	98	99	100	99	97	99	98	99	100	100	99	99	99	100	100	100	99
	B	99	94	97	98	100	99	97	99	98	99	100	100	99	99	99	100	100	100	99
Scaling	+50%	99	98	99	94	95	95	93	94	98	99	99	99	99	99	99	99	100	99	97
	-50%	74	95	84	77	66	71	74	72	73	81	77	79	82	85	84	90	81	85	80
JPEG	1	78	69	73	63	65	64	59	67	63	59	57	58	78	83	80	84	80	82	70
	50	93	95	94	98	99	98	87	80	83	84	89	86	88	88	88	90	89	89	90
	100	99	99	99	100	99	99	98	98	98	99	100	99	99	99	99	99	99	99	99



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Comparison and Generalization Tests

	StarGAN	StyleGAN	StyleGAN2
AutoGAN	65.6	79.5	72.5
FakeSpotter	88	99.1	91.9
EM	90.55	99.48	99.64
CTF (our)	99.9	100	100

Comparison with state-of-the-art methods. Classification of Real images (CelebA and FFHQ) vs. Deepfake images. **ACCURACY VALUES** (%) of each classification task are reported.

	StyleGAN	StarGAN	FF++
Wang	96.3	100	98.2
EM	99	93	98.8
CTF (our)	99.9	99.9	99.9

Comparison with state-of-the-art methods. Classification of Real images (CelebA and FFHQ) vs. Deepfake images. The CTF approach was tested and compared also considering the dataset of Deepfake video's FaceForensics++ (FF++). **AVERAGE PRECISION** values (%) of each classification task are reported.



On the exploitation of DCT-traces in the Generative-AI domain

Introduction and Motivation

- In the age of Artificial Intelligence, deepfakes have emerged as a significant challenge due to their realistic images created using Generative Adversarial Networks (GANs) and Diffusion Models (DMs).
- Distinguishing real from synthetic images has become a priority in Multimedia Forensics, impacting security, information, and privacy.
- Different approaches in the literature (mainly based on deep learning techniques) are able to define not only the nature of the data under analysis (real or deepfake), but also establish the specific model used for creation. But these methods struggle with generalization.



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Introduction and Motivation

- The main contributions of this paper are:
 - One of the first statistical analyses conducted on such a large collection of images generated by both GANs and DMs.
 - A preliminary exploration to determine a differentiating trace between the three classes (real, GANs, DMs) in the DCT domain.



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Proposed Method

The method can be summarized into 3 steps:

- ❶ Extraction of DCT statistics from digital images (real, GAN-generated, DM-generated).
- ❷ Selection of β^{AC} subsets using different techniques ('hand-crafted' and LIME selection).
Training of Machine Learning model using the selected β^{AC} subset and comparing
- ❸ performances in testing phase.

The aim of the research is to identify the most discriminative and robust (to a JPEG compression attacks) subset of DCT statistics.



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Dataset

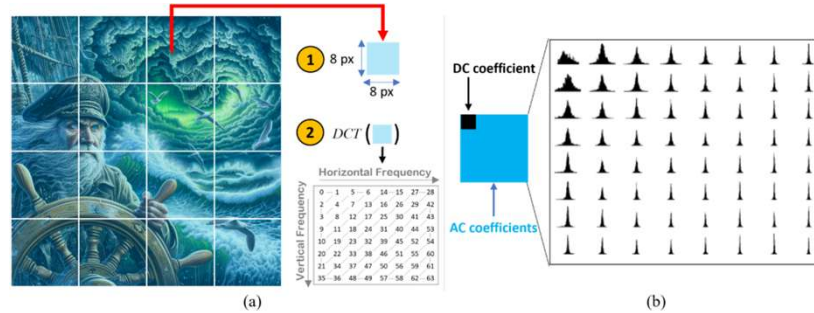


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Extraction of DCT statistics



$$DCT(u, v) = \frac{C(u) \cdot C(v)}{4} \sum_{x=0}^7 \sum_{y=0}^7 f(x, y) \cdot t(x, u) \cdot t(y, v)$$

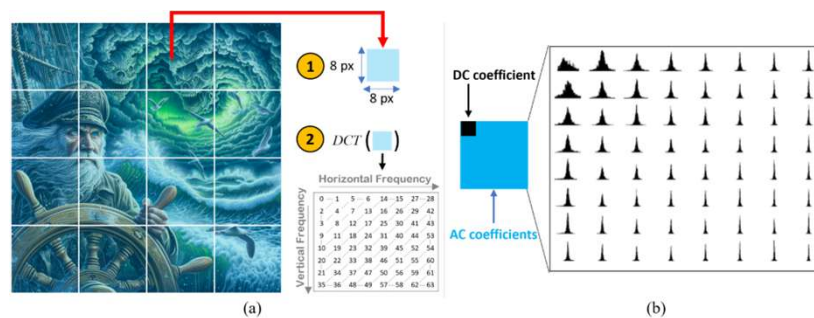
$$\text{where } C(p) = \begin{cases} \frac{1}{\sqrt{2}} & \text{if } p = 0 \\ 1 & \text{otherwise} \end{cases} \text{ and } t(z, \epsilon) = \cos \left[\frac{(2z+1)\epsilon\pi}{16} \right].$$



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Laplacian Distribution



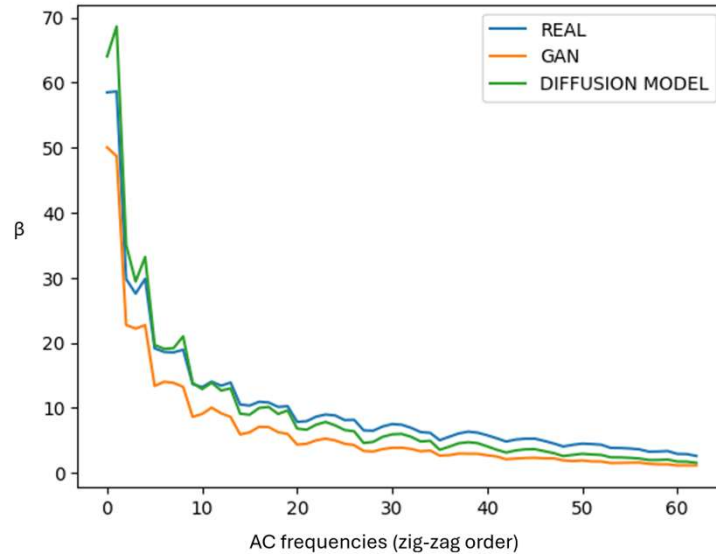
$$P(x|\mu, \beta) = \frac{1}{2\beta} \exp \left(-\frac{|x - \mu|}{\beta} \right)$$



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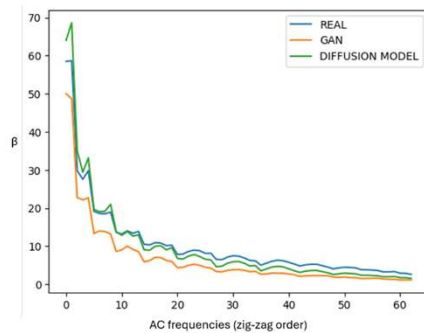


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Selection of β^{AC} subsets



$$F_k = \{\beta_i^{AC} | 1 \leq i \leq k\}$$

$$C_s = \{\beta_i^{AC} | s-1 \leq i \leq s+1\}$$

$$L_t = \{\beta_i^{AC} | 63-t \leq i \leq 63\}$$

with $k, t = 1, \dots, 30$ and $s = 0, \dots, 15$.

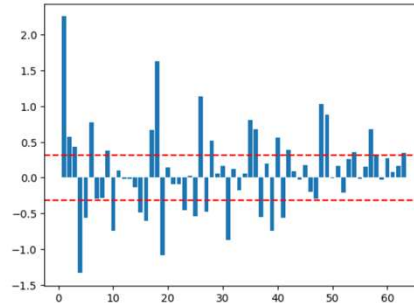
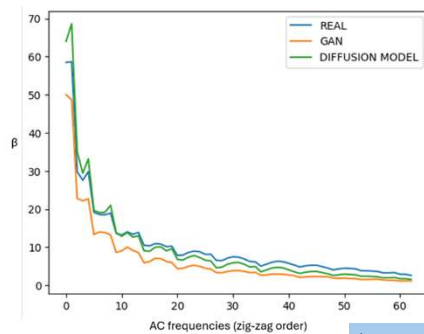


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Selection of β^{AC} subsets



importanza media del coefficiente

$$C_{\beta_i^{AC}}^{avg} = \frac{1}{N} \sum_{j=1}^N c_{\beta_i^{AC}}(p_j) \quad i = 1, \dots, 63$$

contributo dato da LIME

N = numero di predizioni corrette

$$F_k = \{\beta_i^{AC} | 1 \leq i \leq k\}$$

$$C_s = \{\beta_i^{AC} | s-1 \leq i \leq s+1\}$$

$$L_t = \{\beta_i^{AC} | 63-t \leq i \leq 63\}$$

with $k, t = 1, \dots, 30$ and $s = 0, \dots, 15$.

$$POS = \{\beta_i^{AC} : C_{\beta_i^{AC}}^{avg} > 0\}$$

$$ABS = \{\beta_i^{AC} : |C_{\beta_i^{AC}}^{avg}| > Med_j(|C_{\beta_j^{AC}}^{avg}|)\}$$

Contiene coefficienti con contributo positivo

Contiene coefficienti con contributo assoluto sopra la mediana

Interpretazione: coefficienti che mediamente aiutano la classificazione

Interpretazione: coefficienti più "forti" in assoluto, indipendentemente dal segno

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Experiments

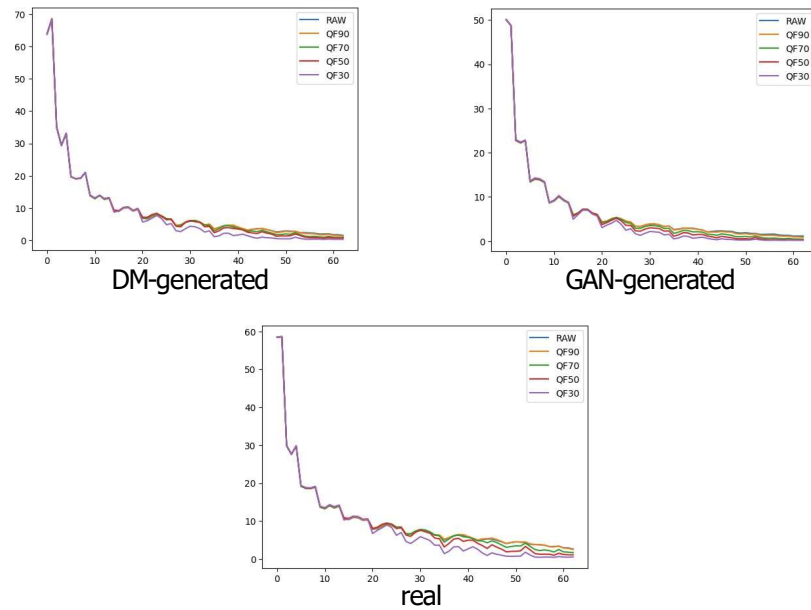
Our study extends from the analysis of the existence of such traces, contained in subsets of β^{AC} coefficients, in RAW images to the study of their persistence after JPEG compression. The experiments conducted are therefore divided into two phases:

- The three machine learning models are trained using different subsets of β^{AC} coefficients extracted from RAW images.
- Models are then tested to infer the nature images using only a proper combination of β^{AC} : if the extracted metrics approach or exceed those obtained from using the full range of coefficients, then we can infer a high discriminative degree of this subset.



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Acc/F1 (%)	K-NN JPEG Compression Test					RANDOM FOREST JPEG Compression Test					GRADIENT BOOSTING JPEG Compression Test				
	RAW	QF90	QF70	QF50	QF30	RAW	QF90	QF70	QF50	QF30	RAW	QF90	QF70	QF50	QF30
β_{AC} ALL	68.76/66.05	68.45/65.65	64.34/61.53	56.46/53.55	47.78/45.47	79.59/77.81	41.75/39.58	33.30/30.23	29.54/26.81	29.44/26.37	82.57/80.81	44.68/44.08	35.10/32.84	32.22/29.26	30.58/26.93
β_{AC} 1:28	64.56/61.87	64.61/62.02	63.82/61.22	63.16/60.13	61.43/58.60	71.13/69.11	69.32/67.42	56.34/55.64	43.63/43.44	34.58/33.94	72.33/70.32	70.86/68.83	57.63/57.09	43.71/44.07	35.01/34.58
β_{AC} 1:29	64.96/62.30	64.86/62.31	63.82/61.26	63.22/60.25	61.17/58.31	71.18/69.18	69.49/67.54	54.91/54.27	41.19/40.80	32.84/31.85	73.03/70.99	71.38/69.30	56.56/56.26	41.49/41.69	33.35/31.91
β_{AC} 1:30	64.80/62.16	64.66/62.12	64.08/61.44	63.17/60.14	60.35/57.39	71.78/69.84	69.67/67.81	54.11/53.55	40.33/39.97	32.39/31.31	73.29/71.25	71.84/69.82	56.23/56.00	41.37/41.75	33.02/31.41
β_{AC} 15:46	69.62/67.08	67.70/65.41	48.09/48.22	32.25/31.78	24.78/23.57	74.10/72.12	53.42/53.20	34.35/31.93	28.54/25.19	29.32/22.98	76.71/74.66	53.31/53.47	36.21/33.96	31.10/26.90	29.68/22.54
β_{AC} 17:45	70.21/67.55	67.86/65.50	47.13/47.40	31.84/31.30	24.00/22.75	74.44/72.45	50.71/50.58	34.28/31.74	29.52/25.50	29.15/21.32	76.49/74.39	53.89/54.01	36.12/33.79	31.32/27.12	29.69/22.58
β_{AC} 18:44	69.65/66.76	66.51/64.10	45.23/45.55	30.42/29.69	23.74/22.25	73.93/71.85	51.46/51.36	33.98/31.52	29.22/25.49	28.83/21.22	76.06/74.00	54.79/54.84	35.51/33.04	31.41/27.50	29.66/23.16
β_{AC} 20:63	72.91/70.04	54.56/54.41	31.63/30.68	26.55/24.94	23.51/21.94	80.10/78.19	40.34/37.98	34.28/31.19	31.73/28.47	29.89/25.98	81.73/79.79	42.34/41.02	35.01/32.18	32.65/29.33	30.26/26.02
β_{AC} 30:63	73.08/70.15	54.06/53.90	31.36/30.23	26.14/24.44	23.54/21.81	80.01/78.08	40.44/38.08	34.53/31.43	31.86/28.63	29.90/26.05	81.54/79.60	42.21/40.93	35.09/32.28	32.31/28.89	30.38/26.27
β_{AC} 31:63	73.45/70.56	53.70/53.53	30.77/29.52	26.09/24.43	23.63/21.78	79.73/77.78	40.40/38.03	34.23/31.10	32.02/28.79	29.91/26.28	81.48/79.57	42.25/40.90	34.77/31.89	32.53/29.12	30.17/25.96
β_{AC} 32:63	73.47/70.57	53.82/53.58	30.74/29.50	26.09/24.46	23.40/21.76	79.86/77.85	40.55/38.26	34.20/31.04	32.14/28.82	29.97/26.50	81.09/79.15	41.68/40.26	35.26/32.16	33.11/29.49	30.68/26.23
β_{AC} 33:63	73.34/70.42	54.31/53.97	31.00/29.79	26.47/24.83	23.67/22.27	79.48/77.45	40.17/37.82	34.30/31.07	32.32/29.10	29.34/26.26	81.26/79.30	42.51/42.38	34.75/31.82	32.93/29.48	30.51/26.78
β_{AC} POS-LIME	67.74/64.77	67.23/64.22	61.77/59.00	53.24/50.81	44.86/42.87	80.05/78.16	41.92/39.85	33.42/30.25	30.23/27.49	29.02/26.06	82.21/80.34	44.62/43.84	34.72/32.00	31.48/28.62	30.64/27.03
β_{AC} ABS-LIME	68.07/65.25	67.51/64.64	64.81/61.91	59.89/56.94	53.05/50.30	78.08/76.03	43.54/41.97	32.98/30.04	29.79/27.14	29.46/26.33	80.68/78.68	45.33/44.45	34.49/32.12	31.42/29.06	30.58/27.10

Accuracy (%) and F1 Score (%) values obtained during the testing phase. Each row reports the experimental results performed for each specific range of β_{AC} coefficients.

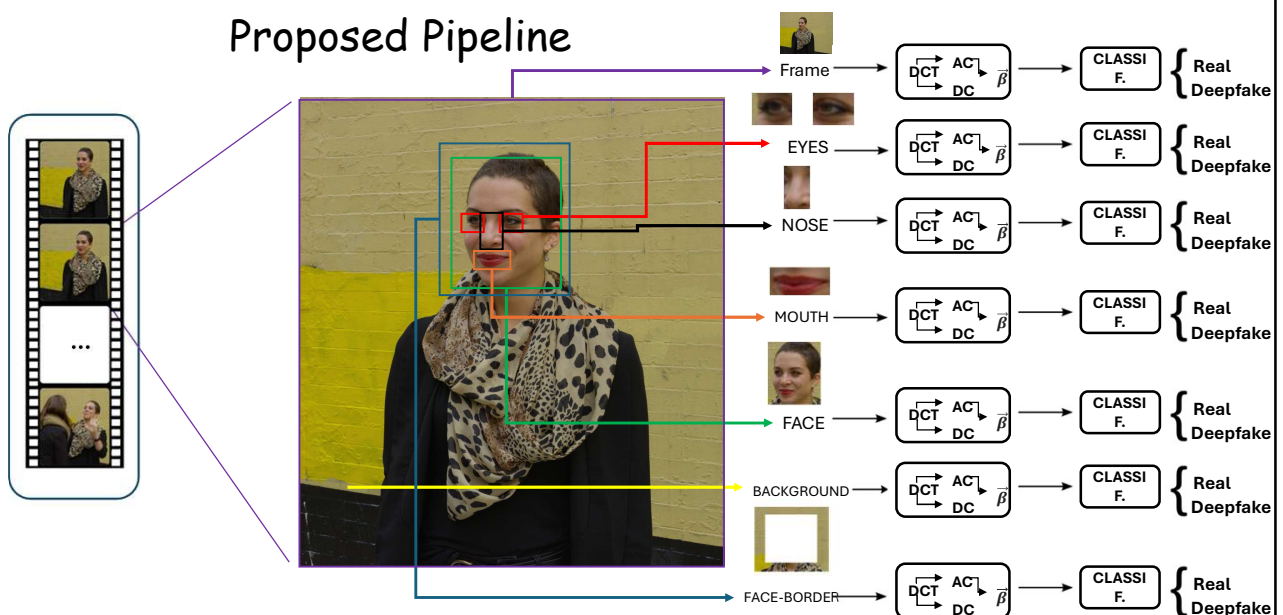
108



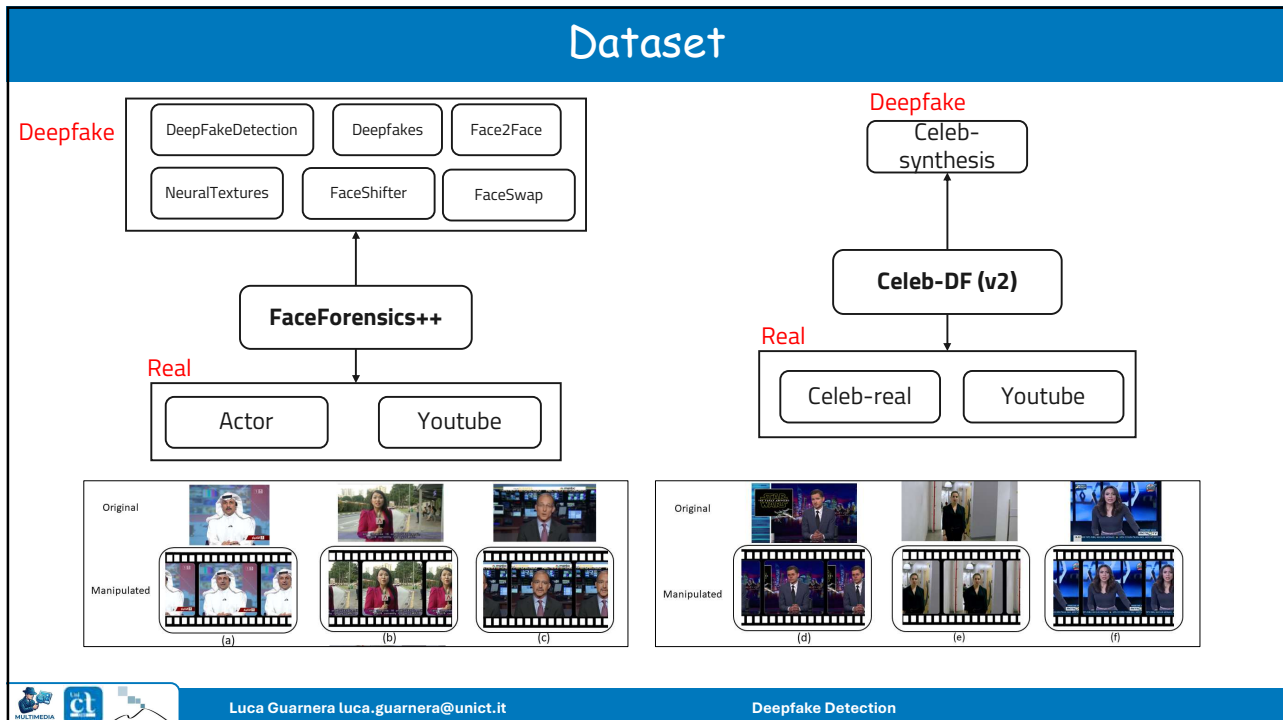
Analisi preliminare nell'utilizzo della DCT per la Video Deepfake Detection

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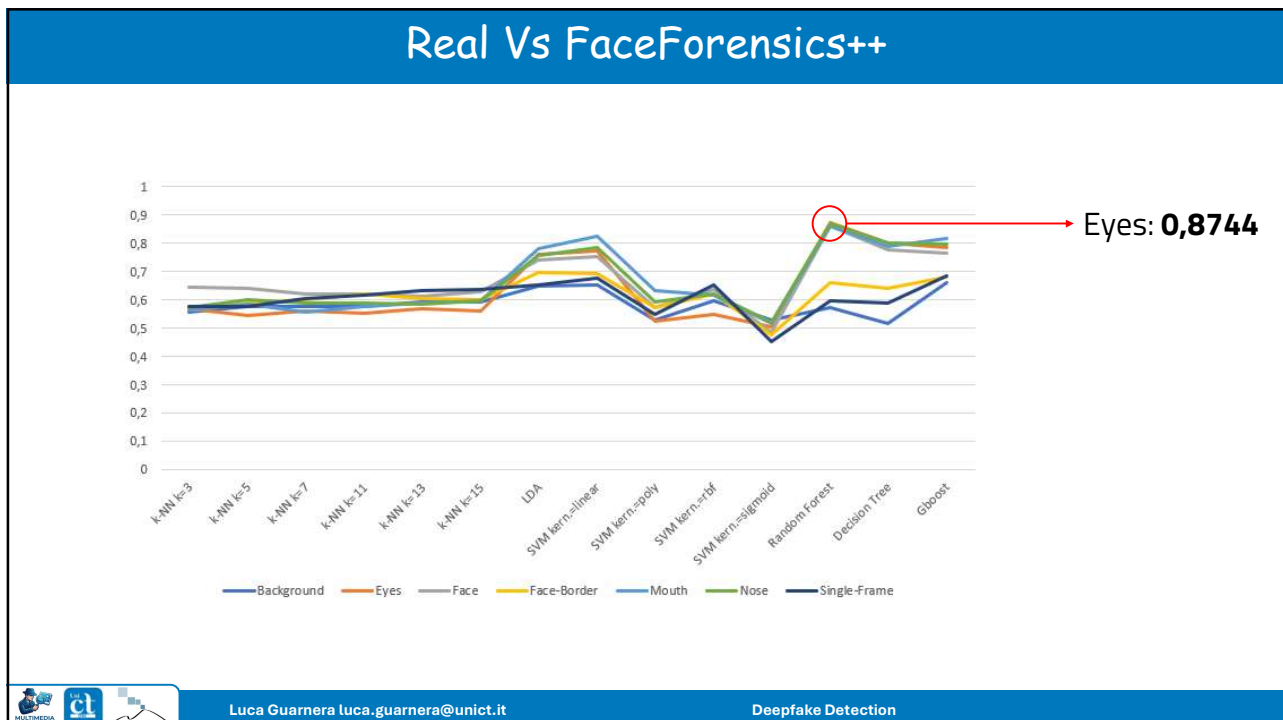
Proposed Pipeline



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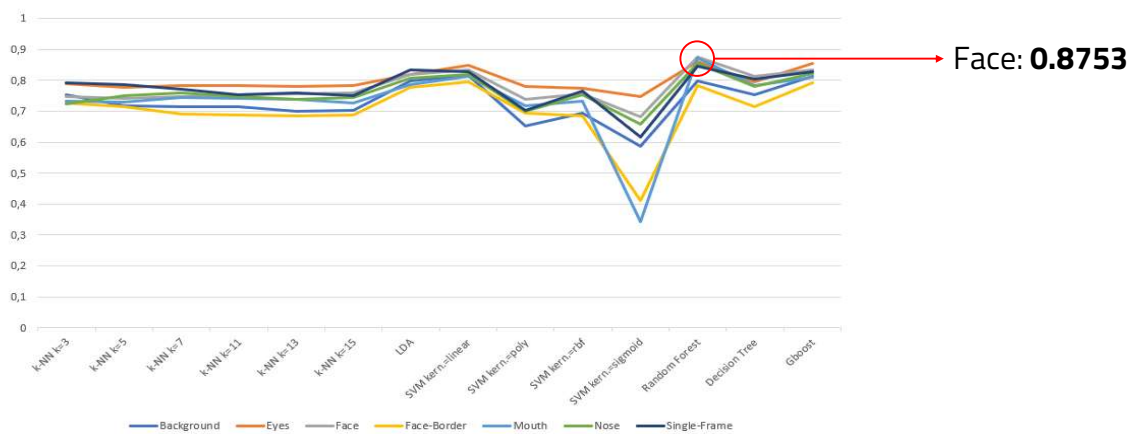


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Real Vs Celeb-DF (v2)



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Real Vs Celeb-DF (v2)-FaceForensics++



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Deepfake Detection

Deep Learning based Approaches

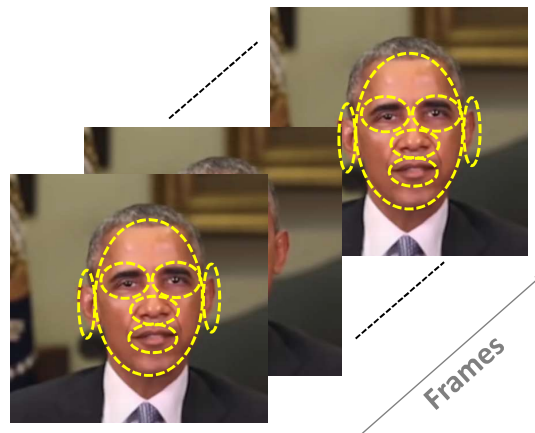
117

Deepfake detection methods

Spatial Artifacts



Spatial and Temporal Artifacts



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CNN-generated images are surprisingly easy to spot... for now

Paper

https://openaccess.thecvf.com/content_CVPR_2020/papers/Wang_CNN-Generated_Images_Are_Surprisingly_Easy_to_Spot..._for_Now_CVPR_2020_paper.pdf

CNN-generated images are surprisingly easy to spot... for now

Sheng-Yu Wang¹ Oliver Wang² Richard Zhang² Andrew Owens^{1,3} Alexei A. Efros¹
UC Berkeley¹ Adobe Research² University of Michigan³



ProGAN [19] StyleGAN [20] BigGAN [7] CycleGAN [40] StarGAN [10] GauGAN [26] CRN [9] IMLE [13] STTD [8] Super-Res [11] DeepFakes [14]

Figure 1: Are CNN-generated images hard to distinguish from real images? We show that a classifier trained to detect images generated by only one CNN (ProGAN, far left) can detect those generated by many other models (remaining columns). Our code and models are available at <https://peterwang512.github.io/CNNDetection/>.

Abstract

In this work we ask whether it is possible to create a “universal” detector for telling apart real images from those generated by a CNN, regardless of architecture or dataset used. To test this, we collect a dataset consisting of fake images generated by 11 different CNN-based image generator models, chosen to span the space of commonly used architectures today (ProGAN, StyleGAN, BigGAN, CycleGAN, StarGAN, GauGAN, DeepFakes, cascaded refinement networks, implicit maximum likelihood es-

are fake [14]. This issue has started to play a significant role in global politics; in one case a video of the president of Gabon that was claimed by opposition to be fake was one factor leading to a failed coup d’etat”. Much of this concern has been directed at specific manipulation techniques, such as “deepfake”-style face replacement [2], and photo-realistic synthetic humans [20]. However, these methods represent only two instances of a broader set of techniques: image synthesis via convolutional neural networks (CNNs). Our goal in this work is to find a general image forensics approach for detecting CNN-generated imagery.

Wang, Sheng-Yu, et al. "CNN-generated images are surprisingly easy to spot... for now." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2020.



Deepfake Detection

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SOTA detection benchmarking

Study	Method	Classifiers	Best Performance	Databases (Generation)
McCloskey and Albright (2018) [55]	GAN-Pipeline Features	SVM	AUC = 70.0%	NIST MFC2018
Wang <i>et al.</i> (2019) [56]	GAN-Pipeline Features	SVM	Acc. = 84.7%	Own (InterFaceGAN, StyleGAN)
Guarnera <i>et al.</i> (2020) [57]	GAN-Pipeline Features	k-NN, SVM, LDA	Acc. = 99.81%	Own (AttGAN, GDWCT, StarGAN, StyleGAN, StyleGAN2)
Nataraj <i>et al.</i> (2019) [58]	Steganalysis Features	CNN	EER = 12.3% [16]	100K-Faces (StyleGAN)
Yu <i>et al.</i> (2019) [59]	Deep Learning Features	CNN	Acc. = 99.5%	Own (ProGAN, SNGAN, CramerGAN, MMDGAN)
Marra <i>et al.</i> (2019) [60]	Deep Learning Features	CNN + Incremental Learning	Acc. = 99.3%	Own (CycleGAN, ProGAN, Glow, StarGAN, StyleGAN)
Dang <i>et al.</i> (2020) [17]	Deep Learning Features	CNN + Attention Mechanism	AUC = 100% EER = 0.1%	DFFD (ProGAN, StyleGAN)
Neves <i>et al.</i> (2020) [16]	Deep Learning Features	CNN	EER = 0.3% EER = 4.5%	100K-Faces (StyleGAN) iFakeFaceDB
Hulzebosch <i>et al.</i> (2020) [61]	Deep Learning Features	CNN, AE	Acc. = 99.8%	Own (StarGAN, Glow, ProGAN, StyleGAN)

Tolosana et al.(2020)



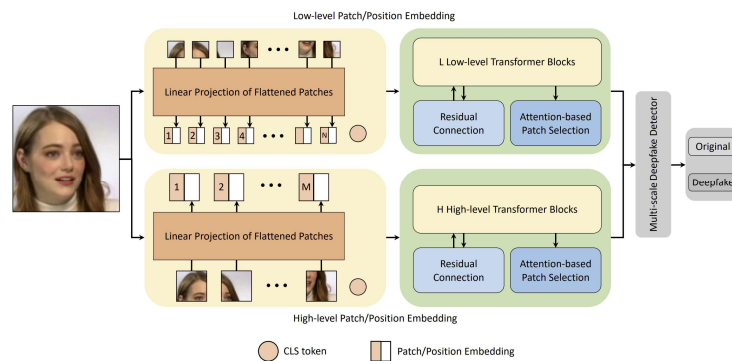
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Deepfake Detection

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DFDT: An End-to-End DeepFake Detection Framework Using Vision Transformer

This work presents DFDT, an end-to-end deepfake detection framework that leverages the unique characteristics of transformer models, for learning hidden traces of perturbations from both local image features and global relationship of pixels at different forgery scales.



Khormali, Aminollah, and Jiann-Shiun Yuan. "DFDT: an end-to-end deepfake detection framework using vision transformer." *Applied Sciences* 12.6 (2022): 2953.



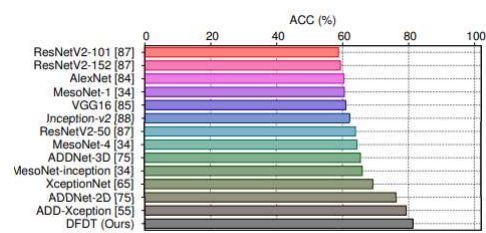
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Deepfake Detection

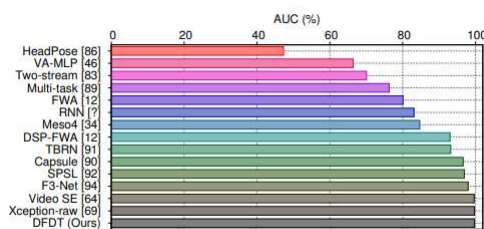
124

Results

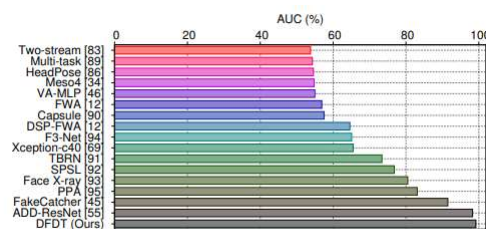
- FaceForensics++ : 99.41%,
- Celeb-DF (V2): 99.31%
- WildDeepfake: 81.35%



(c) WildDeepfake



(a) FaceForensics++



(b) Celeb-DF (V2)



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Deepfake Detection

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State of Art (CVPR 2024)

AVFF: Audio-Visual Feature Fusion for Video Deepfake Detection

Trevine Oorloff, Surya Koppiseti, Nicolò Bonettini, Divyaraj Solanki, Ben Colman, Yaser Yacoob, Ali Shahriyari, Gaurav Bharaj
[pdf] [supp] [bibtex]

Exploiting Style Latent Flows for Generalizing Deepfake Video Detection

Jongwook Choi, Taehoon Kim, Yonghyun Jeong, Seungryul Baek, Jongwon Choi
[pdf] [supp] [arXiv] [bibtex]

Preserving Fairness Generalization in Deepfake Detection

Li Lin, Xinan He, Yan Ju, Xin Wang, Feng Ding, Shu Hu
[pdf] [supp] [arXiv] [bibtex]

Transcending Forgery Specificity with Latent Space Augmentation for Generalizable Deepfake Detection

Zhiyuan Yan, Yuhao Luo, Siwei Lyu, Qingshan Liu, Baoyuan Wu
[pdf] [supp] [arXiv] [bibtex]

Contrastive Learning for DeepFake Classification and Localization via Multi-Label Ranking

Cheng-Yao Hong, Yen-Chi Hsu, Tyngh-Luh Liu
[pdf] [supp] [bibtex]

Rethinking the Up-Sampling Operations in CNN-based Generative Network for Generalizable Deepfake Detection

Chuangchuan Tan, Yao Zhao, Shikui Wei, Guanghua Gu, Ping Liu, Yunchao Wei
[pdf] [arXiv] [bibtex]

LAA-Net: Localized Artifact Attention Network for Quality-Agnostic and Generalizable Deepfake Detection

Dat Nguyen, Nesryne Mejri, Inder Pal Singh, Polina Kuleshova, Marcella Astrid, Anis Kacem, Enjie Ghorbel, Djamila Aouada
[pdf] [supp] [bibtex]



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Deepfake Detection

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LAA-Net: Localized Artifact Attention Network for Quality-Agnostic and Generalizable Deepfake Detection

Dat Nguyen¹, Nesryne Mejri¹, Inder Pal Singh¹, Polina Kuleshova¹, Marcella Astrid¹, Anis Kacem¹, Enjie Ghorbel^{1,2}, Djamila Aouada¹

¹Cvi2, Snt, University Of Luxembourg; ²Cristal Laboratory, National School Of Computer Sciences, University Of Manouba

Abstract

This paper introduces a novel approach for high-quality deepfake detection called Localized Artifact Attention Network (LAA-Net). Existing methods for high-quality deepfake detection are mainly based on a supervised binary classifier coupled with an implicit attention mechanism. As a result, they do not generalize well to unseen manipulations. To handle this issue, two main contributions are made. First, an explicit attention mechanism within a multi-task learning framework is proposed. By combining heatmap-based and self-consistency attention strategies, LAA-Net is forced to focus on a few small artifact-prone vulnerable regions. Second, an Enhanced Feature Pyramid Network (E-FPN) is proposed as a simple and effective mechanism for spreading discriminative low-level features into the final feature output, with the advantage of limiting redundancy. Experiments performed on several benchmarks show the superiority of our approach in terms of Area Under the Curve (AUC) and Average Precision (AP). The code is available at <https://github.com/10Ring/LAA-Net>.

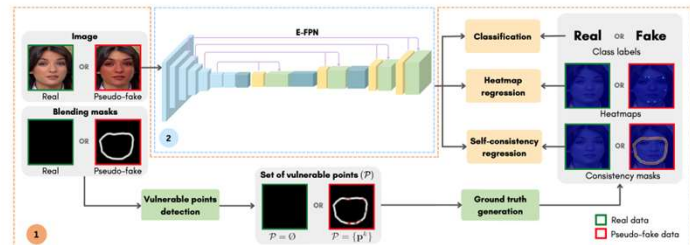


Figure 2. Overview of the proposed LAA-Net approach: it is formed by two components, namely, (1) an explicit attention mechanism based on a multi-task learning framework composed of three branches, i.e., the binary classification branch, the heatmap branch, and the self-consistency branch. The heatmap and self-consistency ground-truth data are generated based on the detected vulnerable points, and (2) an Enhanced Feature Pyramid Networks (E-FPN) that aggregates multi-scale features.

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Deepfake Detection

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AVFF: Audio-Visual Feature Fusion for Video Deepfake Detection

Trevine Oorloff^{1,2}, Surya Koppiseti², Nicolò Bonettini², Divyaraj Solanki², Ben Colman², Yaser Yacoub¹, Ali Shahriyari², Gaurav Bharaj²

¹University of Maryland - College Park; ²Reality Defender Inc.

Abstract

With the rapid growth in deepfake video content, we require improved and generalizable methods to detect them. Most existing detection methods either use uni-modal cues or rely on supervised training to capture the dissonance between the audio and visual modalities. While the former disregards the audio-visual correspondences entirely, the latter predominantly focuses on discerning audio-visual cues within the training corpus, thereby potentially overlooking correspondences that can help detect unseen deepfakes. We present **Audio-Visual Feature Fusion (AVFF)**, a two-stage cross-modal learning method that explicitly captures the correspondence between the audio and visual modalities for improved deepfake detection. The first stage pursues representation learning via self-supervision on real videos to capture the intrinsic audio-visual correspondences. To extract rich cross-modal representations, we use contrastive learning and autoencoding objectives, and introduce a novel audio-visual complementary masking and feature fusion strategy. The learned representations are tuned in the second stage, where deepfake classification is pursued via supervised learning on both real and fake videos. Extensive experiments and analysis suggest that our novel representation learning paradigm is highly discriminative in nature. We report 98.6% accuracy and 99.1% AUC on the FakeAVCeleb dataset, outperforming the current audio-visual state-of-the-art by 14.9% and 9.9%, respectively.

CVPR 2024 – Open Access papers

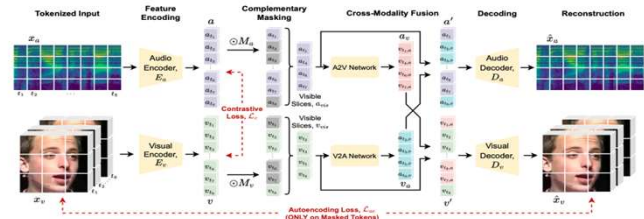


Figure 2. **Audio-Visual Representation Learning Stage.** A real input sample, $x \in \mathcal{D}_r$, with corresponding audio and visual tokens (x_a, x_v) , is split along the temporal dimension, creating K slices, $\{x_{a,t_i}\}_{i=1}^K$ and $\{x_{v,t_i}\}_{i=1}^K$ (illustrated with $K = 8$ in the figure). The temporal slices are then encoded using unimodal transformers, E_a and E_v , to yield feature embeddings a and v . We then complementarily mask 50% of the temporal slices in (a, v) with binary masks (M_a, M_v) . The visible slices of a and v are passed through AZV and V2A networks respectively, to generate cross-modal slices a' and v' . The masked slices of a and v are then replaced with the corresponding slices in a' and v' . The resulting cross-modal fusion representations, a' and v' , are input to unimodal decoders to obtain the audio and visual reconstructions, $\hat{x}_a$ and $\hat{x}_v$. For the learning, we use a dual-objective loss function, which computes the contrastive loss between the audio and visual feature embeddings and the autoencoder loss between the input and the reconstruction of the masked tokens.

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Deepfake Detection

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CVPR 2024 – Workshop on Media Forensics (1/2)

E3: Ensemble of Expert Embedders for Adapting Synthetic Image Detectors to New Generators Using Limited Data

Aref Azizpour, Tai D. Nguyen, Manil Shrestha, Kaidi Xu, Edward Kim, Matthew C. Stamm
[pdf] [arXiv] [bibtex]

An Investigation into the Impact of AI-Powered Image Enhancement on Forensic Facial Recognition

Justin Norman, Hany Farid
[pdf] [bibtex]

Beyond Deepfake Images: Detecting AI-Generated Videos

Danial Samadi Vahdati, Tai D. Nguyen, Aref Azizpour, Matthew C. Stamm
[pdf] [arXiv] [bibtex]

Fusion Transformer with Object Mask Guidance for Image Forgery Analysis

Dimitrios Karageorgiou, Giorgos Kordopatis-Zilos, Symeon Papadopoulos
[pdf] [supp] [arXiv] [bibtex]

Beyond the Screen: Evaluating Deepfake Detectors under Moire Pattern Effects

Razaib Tariq, Minji Heo, Simon S. Woo, Shahroz Tariq
[pdf] [bibtex]

Building Secure and Engaging Video Communication by Using Monitor Illumination

Jun Myeong Choi, Johnathan Leung, Noah Frahm, Max Christman, Gedas Bertasius, Roni Sengupta
[pdf] [supp] [bibtex]

StampOne: Addressing Frequency Balance in Printer-proof Steganography

Farhad Shadmand, Iuri Medvedev, Luiz Schirmer, João Marcos, Nuno Gonçalves
[pdf] [supp] [bibtex]

Finding AI-Generated Faces in the Wild

Gonzalo J. Aniano Porcile, Jack Gindi, Shivansh Mundra, James R. Verbus, Hany Farid
[pdf] [arXiv] [bibtex]

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Beyond Deepfake Images: Detecting AI-Generated Videos

Danial Samadi Vahdati, Tai D. Nguyen, Aref Azizpour, Matthew C. Stamm
Drexel University, Philadelphia, PA, USA

CVPR 2024 – 8th Workshop on Media Forensics

Abstract

Recent advances in generative AI have led to the development of techniques to generate visually realistic synthetic video. While a number of techniques have been developed to detect AI-generated synthetic images, in this paper we show that synthetic image detectors are unable to detect synthetic videos. We demonstrate that this is because synthetic video generators introduce substantially different traces than those left by image generators. Despite this, we show that synthetic video traces can be learned, and used to perform reliable synthetic video detection or generator source attribution even after H.264 re-compression. Furthermore, we demonstrate that while detecting videos from new generators through zero-shot transferability is challenging, accurate detection of videos from a new generator can be achieved through few-shot learning.

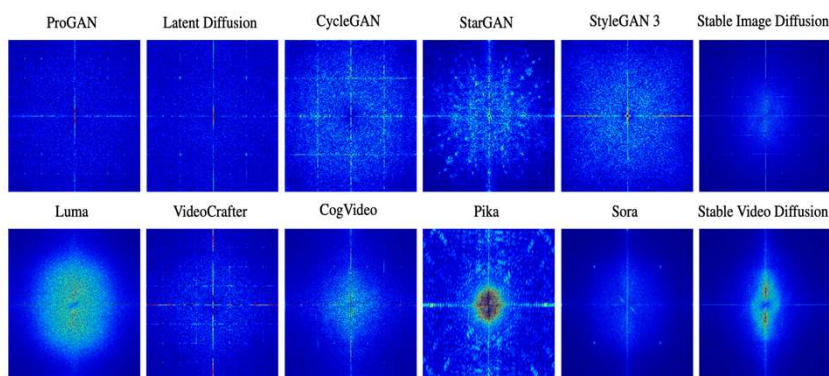


Figure 3. Fourier transform analysis of the forensic traces extracted from different synthetic image and video generators.

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Deepfake Detection

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CVPR 2024 – Workshop on Media Forensics (2/2)

Raising the Bar of AI-generated Image Detection with CLIP

Davide Cozzolino, Giovanni Poggi, Riccardo Corvi, Matthias Nießner, Luisa Verdoliva
[pdf] [supp] [arXiv] [bibtex]

Audio Transformer for Synthetic Speech Detection via Multi-Formant Analysis

Luca Cuccovillo, Milica Gerhardt, Patrick Aichroth
[pdf] [bibtex]

FairSSD: Understanding Bias in Synthetic Speech Detectors

Amit Kumar Singh Yadav, Kratika Bhagatani, Davide Salvi, Paolo Bestagini, Edward J. Delp
[pdf] [supp] [arXiv] [bibtex]

Can ChatGPT Detect DeepFakes? A Study of Using Multimodal Large Language Models for Media Forensics

Shan Jia, Reilin Lyu, Kangran Zhao, Yize Chen, Zhiyuan Yan, Yan Ju, Chuanbo Hu, Xin Li, Baoyuan Wu, Siwei Lyu
[pdf] [arXiv] [bibtex]

Audio Provenance Analysis in Heterogeneous Media Sets

Milica Gerhardt, Luca Cuccovillo, Patrick Aichroth
[pdf] [bibtex]

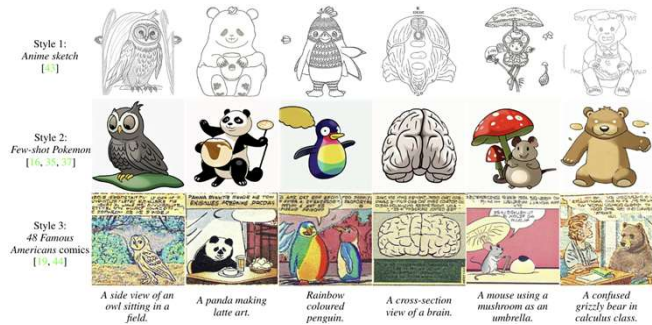
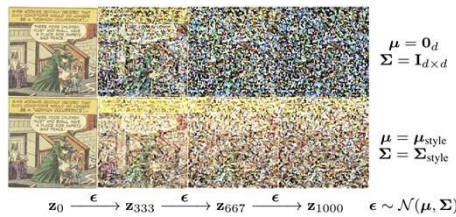
Lost in Translation: Lip-Sync Deepfake Detection from Audio-Video Mismatch

Matyas Bohacek, Hany Farid
[pdf] [bibtex]



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Some (new) ideas: GAN Fingerprint in the latent space



Everaert, Martin Nicolas, et al. "Diffusion in style." *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023.

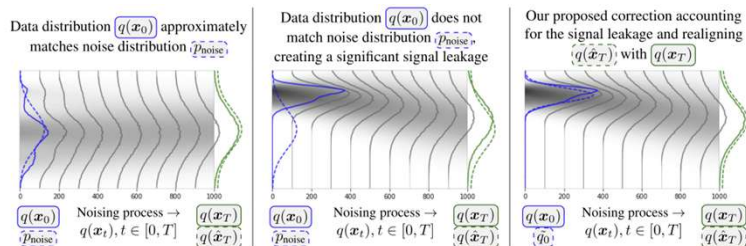


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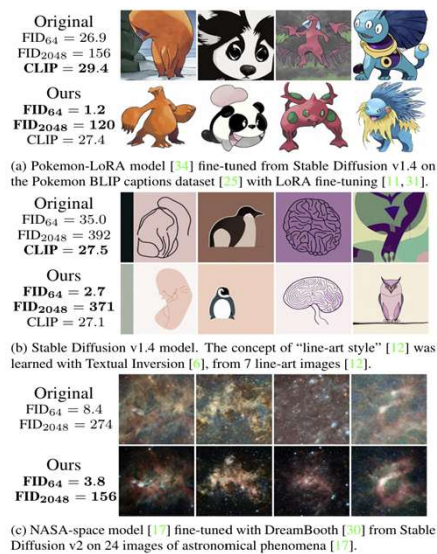
Deepfake Detection

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Exploiting the signal-leak bias in diffusion models



Everaert, Martin Nicolas, et al. "Exploiting the signal-leak bias in diffusion models." *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2024.



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Deepfake Detection

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DeepFeatureX Net: Deep Features eXtractors based Network for discriminating synthetic from real images

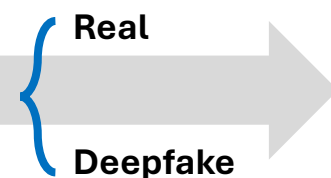
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La maggior parte dei detector esistenti ...

- Inizialmente tutti i metodi di Deepfake Detection erano basati sull'uso di architetture all'avanguardia



Base architecture
(ResNet, EfficientNet, ...)

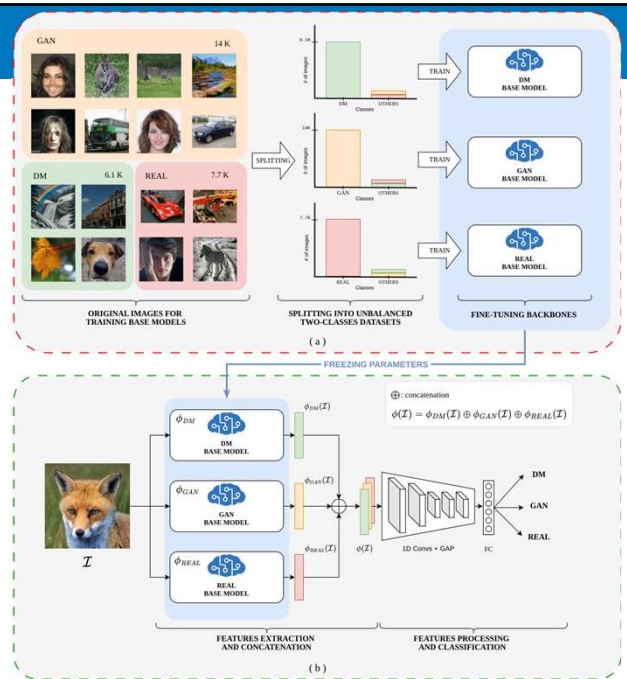


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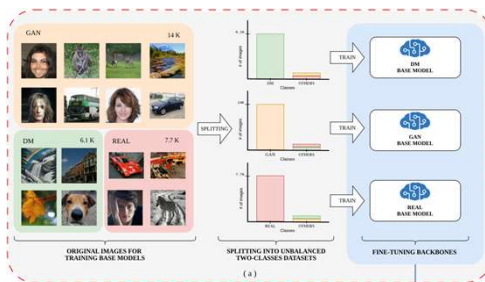
DeepFeatureX Net

The following CNNs were used as backbone:

- DenseNet 121, DenseNet 161, DenseNet 169, DenseNet 201, EfficientNet b0, EfficientNet b4, ResNet 18, ResNet 34, ResNet 50, ResNet 101, ResNet 152, ResNeXt 101, ViT b16, ViT b32.
- All backbones have been pretrained on the Imagenet dataset



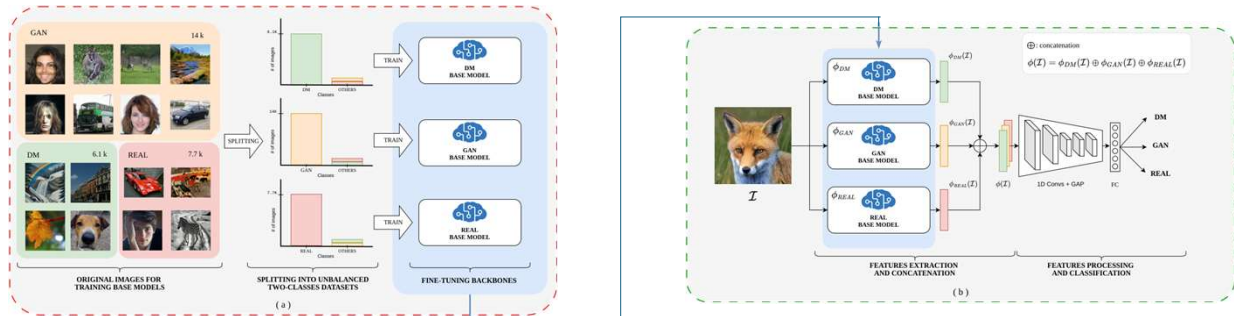
Training of Base Models



Backbone	DM Base Model				GAN Base Model				REAL Base Model			
	Acc	Rec	Pre	F1	Acc	Rec	Pre	F1	Acc	Rec	Pre	F1
DenseNet 121	76.34	99.00	47.60	64.29	92.34	99.25	87.64	93.08	73.17	99.08	49.73	66.22
DenseNet 161	83.96	98.64	57.40	72.57	94.62	99.45	91.03	95.05	77.39	99.21	54.04	69.97
DenseNet 169	78.83	99.16	50.40	66.83	93.86	99.32	89.91	94.38	71.20	99.29	47.95	64.67
DenseNet 201	79.11	98.90	50.75	67.08	92.61	99.37	87.96	93.32	72.97	99.29	49.54	66.10
EfficientNet b0	85.74	97.09	60.50	74.55	88.77	98.81	82.87	90.14	77.91	97.56	54.70	70.10
EfficientNet b4	78.00	97.47	49.43	65.59	87.14	98.74	80.78	88.86	74.31	97.27	50.84	66.78
ResNet 18	76.64	98.03	47.90	64.36	84.28	99.06	77.16	86.75	65.14	99.03	43.17	60.13
ResNet 34	77.29	98.12	48.62	65.02	83.33	99.40	75.93	86.10	66.75	98.82	44.33	61.21
ResNet 50	78.14	98.61	49.59	66.00	90.34	99.14	84.82	91.42	70.79	99.13	47.59	64.31
ResNet 101	77.20	99.00	48.53	65.13	90.85	98.87	85.71	91.82	69.10	99.08	46.17	62.99
ResNet 152	76.48	99.00	47.75	64.43	93.42	99.32	89.23	94.00	70.59	98.92	47.41	64.10
ResNeXt 101	75.38	98.58	46.60	63.28	93.82	98.40	90.53	94.30	66.26	98.66	43.96	60.82
ViT b16	76.53	98.58	47.80	64.38	83.58	99.69	76.11	86.32	68.45	99.24	45.67	62.55
ViT b32	74.05	96.92	45.20	61.65	80.83	98.99	73.39	84.29	60.44	99.61	40.13	57.21

Percentage values of the metrics Accuracy, Recall, Precision, and F1 Score obtained by testing the Base Model to the binary classification between the predominant class and the class 'others'

Step 2



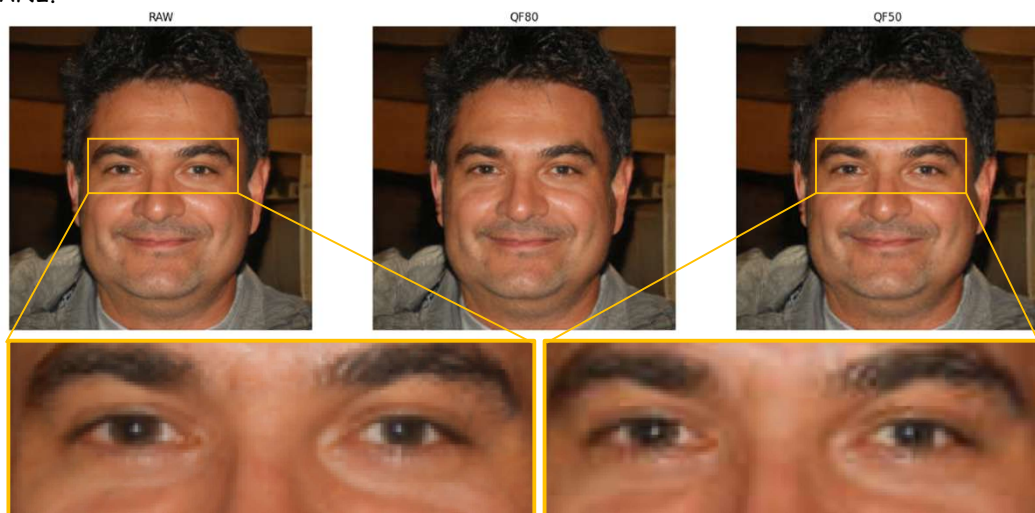
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Robustness Test

- Image variation as JPEG compression Quality Factor decreases. On the left raw image, at center JPEG compressed image at Quality Factor 80, and on the right the image at QF 50. Image generated by StyleGAN2.



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Deepfake Detection

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Test: Real Vs GAN Vs DM

Base Model backbone	Multi-class - Accuracy / F1-Score (%)					
	RAW	JPEG-Compression				
		QF90	QF80	QF70	QF60	QF50
DenseNet 121	91.39/91.35	89.34/89.26	87.19/87.11	84.31/84.26	81.17/81.24	79.19/79.33
DenseNet 161	93.30/93.21	91.72/92.07	88.21/88.04	83.88/84.06	77.78/78.21	74.06/74.53
DenseNet 169	91.11/91.14	89.36/89.36	86.99/87.04	82.99/83.17	77.92/78.31	75.46/75.92
DenseNet 201	92.31/92.30	90.70/90.67	87.78/87.76	84.76/84.76	81.98/82.01	79.78/79.87
EfficientNet b0	88.62/88.66	86.19/86.26	84.74/84.83	82.95/83.05	80.45/80.56	78.98/79.11
EfficientNet b4	84.39/84.51	83.00/83.16	81.59/81.75	80.28/80.45	78.48/78.66	78.42/78.61
ResNet 18	85.22/85.23	84.64/84.64	83.61/83.61	82.72/82.71	80.65/80.68	79.77/79.77
ResNet 34	85.98/86.00	85.37/85.37	84.25/84.23	82.82/82.75	80.37/80.30	79.12/79.02
ResNet 50	89.63/89.62	88.01/87.99	86.07/86.06	83.94/83.95	80.90/80.96	78.85/78.93
ResNet 101	90.96/90.97	89.85/89.84	87.90/87.88	86.53/86.52	83.27/84.32	82.97/83.05
ResNet 152	91.10/91.11	89.57/89.57	87.53/87.54	85.26/85.30	82.18/82.26	80.63/80.72
ResNeXt 101	89.26/89.28	87.89/87.91	87.02/87.04	85.60/85.67	83.71/84.83	82.76/82.61
ViT b16	88.09/88.11	86.96/87.03	85.91/85.99	84.33/84.45	82.80/82.69	79.48/79.69
ViT b32	81.25/81.24	81.23/81.22	81.12/81.07	81.15/81.11	80.73/80.68	80.14/80.08

Percentage values of Accuracy and F1 Score obtained during the testing phase in three-class classification (GAN vs. DM vs. real) at backbone variation



Test: Real Vs Deepfake

Base Model backbone	Binary - Accuracy / F1-Score (%)					
	RAW	JPEG-Compression				
		QF90	QF80	QF70	QF60	QF50
DenseNet 121	92.50/92.23	91.64/91.35	90.67/90.48	87.44/88.55	85.86/86.44	83.85/84.89
DenseNet 161	93.83/93.67	92.74/92.56	90.17/90.16	88.34/88.84	82.75/84.28	79.65/82.15
DenseNet 169	92.10/91.92	90.95/90.69	89.01/88.92	86.21/86.58	82.64/83.79	80.48/82.12
DenseNet 201	93.30/93.03	91.30/90.97	89.45/89.06	88.09/87.81	85.78/85.75	83.78/83.89
EfficientNet b0	89.36/89.01	87.72/87.35	85.86/85.48	84.96/84.57	82.69/82.59	81.57/81.39
EfficientNet b4	86.67/86.56	86.03/86.11	84.58/84.91	84.32/84.67	82.14/82.73	82.26/82.74
ResNet 18	85.62/84.89	85.71/84.96	84.35/83.49	84.05/82.99	81.92/80.84	81.55/80.21
ResNet 34	87.00/86.37	86.99/86.28	85.25/84.28	83.56/82.16	81.58/79.89	79.57/77.48
ResNet 50	90.27/89.83	89.10/88.53	87.12/86.41	85.48/84.71	82.92/82.21	80.24/79.24
ResNet 101	91.59/91.37	91.01/90.68	89.10/88.60	87.72/87.17	85.33/84.89	84.02/83.63
ResNet 152	91.71/91.38	90.44/89.93	88.57/88.03	86.79/86.21	84.60/84.16	82.72/82.20
ResNeXt 101	90.86/90.67	90.50/90.25	89.12/88.93	88.09/87.93	86.54/86.61	86.11/86.13
ViT b16	89.49/89.27	89.75/89.68	88.40/88.43	87.19/87.38	85.36/85.71	82.25/82.59
ViT b32	82.31/81.15	82.63/81.45	82.55/81.38	82.24/81.09	82.33/80.96	81.66/80.20

Percentage values of Accuracy and F1 Score obtained during the testing phase in binary classification (Deepfake vs real) of the model when the backbone varies. Values were obtained considering DM and GAN predictions as Deepfake.



Comparison with S.O.T.A. in generalization

Dataset	Tipo di immagini fake		$\mathcal{T}_G^i$	$\mathcal{T}_G^o$	$\mathcal{T}_G^{i/o}$	$\mathcal{T}_D^i$	$\mathcal{T}_D^o$	$\mathcal{T}_D^{i/o}$	$\mathcal{T}_{G/D}^i$	$\mathcal{T}_{G/D}^o$	$\mathcal{T}_{G/D}^{i/o}$	
T_G^i	GAN viste nel training	Baselines	DenseNet 121	56.57	74.02	66.96	72.07	48.20	58.68	60.58	64.23	63.13
			DenseNet 161	53.56	73.97	66.21	69.19	48.12	56.93	58.55	64.51	61.98
T_G^o	GAN non viste nel training		DenseNet 169	52.03	66.73	61.10	65.25	43.98	52.61	56.80	57.93	57.51
			DenseNet 201	55.93	70.03	64.14	67.39	48.70	56.10	60.71	62.35	62.72
			EfficientNet b0	49.58	71.81	62.99	69.22	45.12	55.19	55.78	61.67	59.85
			EfficientNet b4	50.37	68.64	61.36	69.42	46.60	55.92	56.75	60.98	60.00
$T_G^{i/o}$	GAN viste + non viste		ResNet 18	63.77	68.89	66.65	66.23	55.03	58.30	63.70	62.30	62.82
T_D^i	Diffusion viste nel training		ResNet 34	53.87	70.03	63.25	65.48	48.55	54.76	57.31	61.84	60.98
			ResNet 50	59.58	73.13	67.95	67.89	53.38	58.50	62.10	65.01	63.90
T_D^o	Diffusion non viste nel training		ResNet 101	60.35	68.08	65.40	72.12	56.45	59.68	64.21	63.62	63.20
			ResNet 152	53.94	68.61	61.84	63.90	50.15	55.27	55.27	61.51	60.00
			ResNeXt 101	54.35	67.42	62.86	74.18	50.23	59.57	61.19	61.06	61.87
$T_D^{i/o}$	Diffusion viste + non viste		ViT b16	65.81	73.31	69.46	68.59	52.69	58.30	66.62	64.53	62.72
			ViT b32	54.07	61.91	58.87	60.34	41.87	47.92	56.22	54.25	57.38
$T_{G/D}^i$	GAN + Diffusion viste	SOTA	Gandhi2020 [16]	52.30	50.79	51.71	49.91	50.86	50.34	51.54	50.57	51.06
			Wang2020 [50]	62.41	53.18	57.87	50.13	50.93	50.44	58.26	52.14	54.86
$T_{G/D}^o$	GAN + Diffusion non viste		Arshed2024 [1]	47.46	47.65	48.54	52.69	50.00	51.04	49.89	48.94	52.20
			Guarnera2024 [20]	55.00	55.63	56.23	54.11	45.98	49.97	56.07	52.21	57.17
$T_{G/D}^{i/o}$	GAN + Diffusion viste e non viste		Our	64.74	72.47	69.89	68.09	60.82	59.96	66.06	65.02	64.39

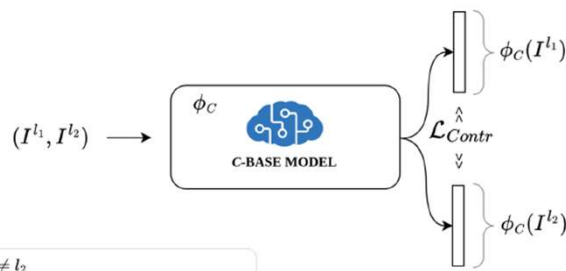


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Basic models: a Siamese neural network configuration



$$l_1 \neq l_2$$

$$l_i, C \in \{DM, GAN, REAL\}, i = 1, 2$$

Contrastive Loss $\mathcal{L}_{Contr}(I_i, I_j, l_{ij}) = \frac{1}{2}(1 - l_{ij})d(I_i, I_j)^2 + \frac{1}{2}l_{ij} \max[0, m - d(I_i, I_j)]^2$

$$d(I_i, I_j) = \|\Phi(I_i) - \Phi(I_j)\|_2$$

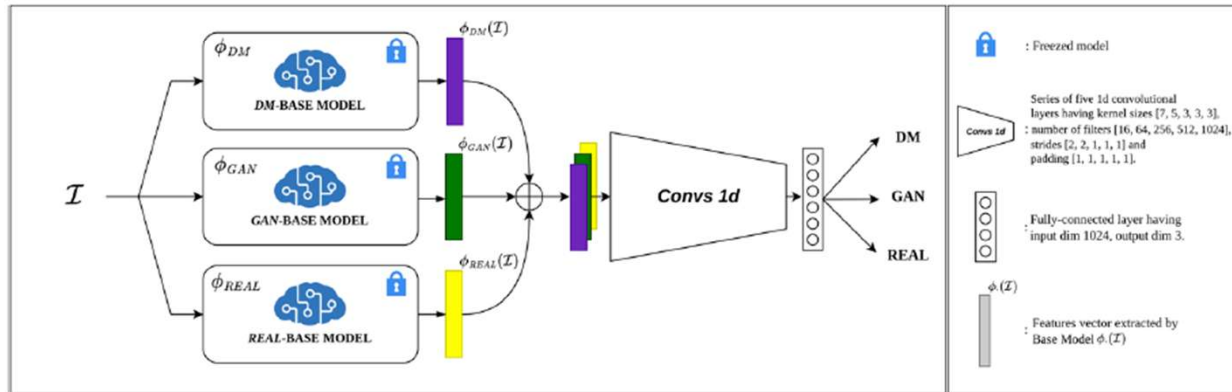


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DeepFeatureX-SN

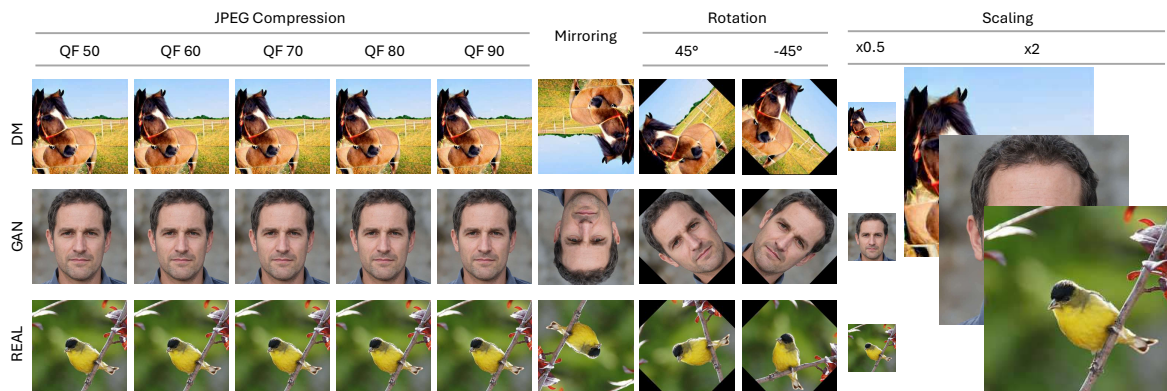


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Dataset



INFERENCE AND ROBUSTNESS TESTS

Cls type	RAW	JPEG Compression					Mirroring	Rotation		Resize	
		QF90	QF80	QF70	QF60	QF50		+45	-45	x2	x0.5
Three-class	97.29	95.17	88.53	74.73	66.43	63.56	95.36	81.12	78.73	66.60	92.08
Binary	97.70	96.48	90.65	78.39	71.91	69.85	96.18	84.52	81.47	61.11	90.19



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Generalization Test

GENERALIZATION TESTS

	$\mathcal{T}_G^i$	$\mathcal{T}_G^o$	$\mathcal{T}_G^{i/o}$	$\mathcal{T}_D^i$	$\mathcal{T}_D^o$	$\mathcal{T}_D^{i/o}$	$\mathcal{T}_{G/D}^i$	$\mathcal{T}_{G/D}^o$	$\mathcal{T}_{G/D}^{i/o}$	Avg.
Gandhi2020 [51]	52.30	50.79	51.71	49.91	50.86	50.34	51.54	50.57	51.06	51.01
Wang2020 [14]	62.41	53.18	57.87	50.13	50.93	50.44	58.26	52.14	54.86	54.47
Arshed2024 [52]	47.46	47.65	48.54	52.69	50.00	51.04	49.89	48.94	52.20	49.82
Guarnera2024 [15]	55.00	55.63	56.23	54.11	45.98	49.97	56.07	52.21	57.17	53.59
Pontorno2024 [53]	65.84	72.47	69.89	68.09	60.82	59.96	66.06	65.02	64.39	65.84
DeepFeatureX-SN (Ours)	66.30	74.45	71.10	72.93	57.24	63.43	66.67	67.30	67.24	67.40

Bold values indicate significance, highlighting the best results

The tests distinguished between images generated from architectures seen in the training phase, but with different initial conditions (superscript i), and images generated from architectures never seen before (superscript o), and mixed (superscript i/o). Furthermore, the tests distinguished between using only images generated by GANs (G-index), those by DMs (D-index), and mixed (G/D-index)



T-SNE Representation

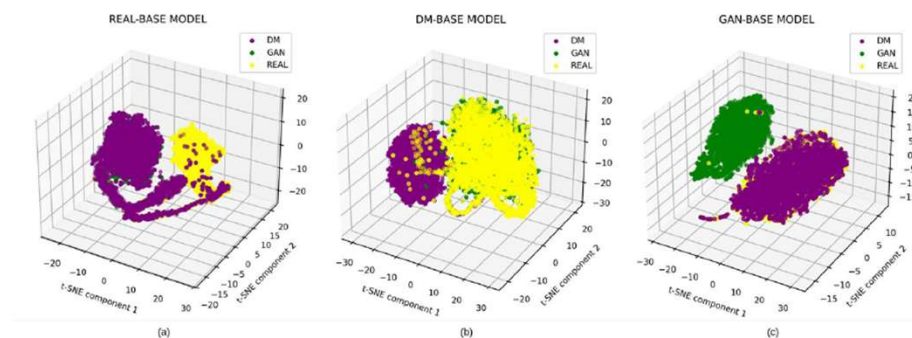


Fig. 5 Spatial representation of features extracted from Base Models: (a) REAL-Base Model, (b) DM-Base Model, (c) GAN-Base Model. The plots were produced using the t-SNE algorithm with dimension 3



Some References

- Amerini, Irene, et al. "Deepfake media forensics: Status and future challenges." *Journal of Imaging* 11.3 (2025): 73.
- Guarnera, Luca, Oliver Giudice, and Sebastiano Battiato. "Fighting deepfake by exposing the convolutional traces on images." *IEEE access* 8 (2020): 165085-165098.
- Giudice, Oliver, Luca Guarnera, and Sebastiano Battiato. "Fighting deepfakes by detecting gan dct anomalies." *Journal of Imaging* 7.8 (2021): 128.
- Pontorno, Orazio, Luca Guarnera, and Sebastiano Battiato. "On the exploitation of DCT-traces in the generative-AI domain." *2024 IEEE international conference on image processing (ICIP)*. IEEE, 2024.
- Pontorno, Orazio, Luca Guarnera, and Sebastiano Battiato. "Deepfeaturex net: Deep features extractors based network for discriminating synthetic from real images." *International Conference on Pattern Recognition*. Cham: Springer Nature Switzerland, 2024.
- Pontorno, Orazio, Luca Guarnera, and Sebastiano Battiato. "DeepFeatureX-SN: Generalization of deepfake detection via contrastive learning." *Multimedia Tools and Applications* (2025): 1-20.



Conclusioni

- Esistono tanti metodi che fanno Deepfake Detection
 - Approcci basati su ML
 - Approcci basati su DL
- Problemi Aperti:
 - Generalizzazione «in the wild»
 - Robustezza in scenari reali
 - Vincoli sulla semantica
 - Interpretabilità e Spiegabilità





Deepfake Detection

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